3 Chapter

Interpolation

3.1 INTRODUCTION

The statement

$$y = f(x), \qquad x_0 \le x \le x_n$$

means: corresponding to every value of x in the range $x_0 \le x \le x_n$, there exists one or more values of y. Assuming that $f(x)$ is single-valued and continuous and that it is known explicitly, then the values of $f(x)$ corresponding to certain given values of x, say $x_0, x_1, \ldots, x_n$ can easily be computed and tabulated. The central problem of numerical analysis is the converse one: Given the set of tabular values $(x_0, y_0), (x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ satisfying the relation $y = f(x)$ where the explicit nature of $f(x)$ is not known, it is required to find a simpler function, say $\phi(x)$, such that $f(x)$ and $\phi(x)$ *agree at the set of tabulated points*. Such a process is called *interpolation*. If $\phi(x)$ is a polynomial, then the process is called *polynomial interpolation* and $\phi(x)$ is called the *interpolating polynomial*. Similarly, different types of interpolation arise depending on whether $\phi(x)$ is a finite trigonometric series, series of Bessel functions, etc. In this chapter, we shall be concerned with polynomial interpolation only. As a justification for the approximation of an unknown function by means of a polynomial, we state here, without proof, a famous theorem due to Weierstrass (1885): if $f(x)$ is continuous in $x_0 \le x \le x_n$, then given any $\varepsilon > 0$, there exists a polynomial $P(x)$ such that

$$|f(x) - P(x)| < \varepsilon, \quad \text{for all } x \text{ in } (x_0, x_n).$$

This means that it is possible to find a polynomial $P(x)$ whose graph remains within the region bounded by $y = f(x) - \varepsilon$ and $y = f(x) + \varepsilon$ for all x between x_0 and x_n, however small ε may be.

When approximating a given function $f(x)$ by means of polynomial $\phi(x)$, one may be tempted to ask: (i) How should the closeness of the approximation be measured? and (ii) What is the criterion to decide the best polynomial approximation to the function? Answers to these questions, important though they are for the practical problem of interpolation, are outside the scope of this book and will not be attempted here. We will, however, derive in the next section a formula for finding the error associated with the approximation of a tabulated function by means of a polynomial.

3.2 ERRORS IN POLYNOMIAL INTERPOLATION

Let the function $y(x)$, defined by the $(n + 1)$ points (x_i, y_i), $i = 0, 1, 2, \ldots, n$, be continuous and differentiable $(n + 1)$ times, and let $y(x)$ be approximated by a polynomial $\phi_n(x)$ of degree not exceeding n such that

$$\phi_n(x_i) = y_i, \quad i = 0, 1, 2, \ldots, n \tag{3.1}$$

If we now use $\phi_n(x)$ to obtain approximate values of $y(x)$ at some points other than those defined by Eq. (3.1), what would be the accuracy of this approximation? Since the expression $y(x) - \phi_n(x)$ vanishes for $x = x_0, x_1, \ldots, x_n$, we put

$$y(x) - \phi_n(x) = L\Pi_{n+1}(x), \tag{3.2}$$

where

$$\Pi_{n+1}(x) = (x - x_0)(x - x_1) \ldots (x - x_n) \tag{3.3}$$

and L is to be determined such that Eq. (3.2) holds for any intermediate value of x, say $x = x'$, $x_0 < x' < x_n$. Clearly,

$$L = \frac{y(x') - \phi_n(x')}{\Pi_{n+1}(x')}. \tag{3.4}$$

We construct a function $F(x)$ such that

$$F(x) = y(x) - \phi_n(x) - L\Pi_{n+1}(x), \tag{3.5}$$

where L is given by Eq. (3.4) above,

It is clear that

$$F(x_0) = F(x_1) = \cdots = F(x_n) = F(x') = 0,$$

that is, $F(x)$ vanishes $(n + 2)$ times in the interval $x_0 \le x \le x_n$; consequently, by the repeated application of Rolle's theorem (see Theorem 1.3, Section 1.2), $F'(x)$ must vanish $(n + 1)$ times, $F''(x)$ must vanish n times, etc., in the interval $x_0 \le x \le x_n$. In particular, $F^{(n+1)}(x)$ must vanish once in the interval.

Let this point be given by $x = \xi$, $x_0 < \xi < x_n$. On differentiating Eq. (3.5) $(n + 1)$ times with respect to x and putting $x = \xi$, we obtain

$$0 = y^{(n+1)}(\xi) - L(n + 1)!$$

so that

$$L = \frac{y^{(n+1)}(\xi)}{(n+1)!}. \tag{3.6}$$

Comparison of Eqs. (3.4) and (3.6) yields the results

$$y(x') - \phi_n(x') = \frac{y^{(n+1)}(\xi)}{(n+1)!}\Pi_{n+1}(x').$$

Dropping the prime on x', we obtain

$$y(x) - \phi_n(x) = \frac{\Pi_{n+1}(x)}{(n+1)!}y^{(n+1)}(\xi), \quad x_0 < \xi < x_n, \tag{3.7}$$

which is the required expression for the error. Since $y(x)$ is, generally, unknown and hence we do not have any information concerning $y^{(n+1)}(x)$, formula (3.7) is almost useless in practical computations. On the other hand, it is extremely useful in theoretical work in different branches of numerical analysis. In particular, we will use it to determine errors in Newton's interpolating formulae which will be discussed in Section 3.6.

3.3 FINITE DIFFERENCES

Assume that we have a table of values (x_i, y_i), $i = 0, 1, 2, \ldots, n$ of any function $y = f(x)$, the values of x being equally spaced, i.e., $x_i = x_0 + ih$, $i = 0, 1, 2, \ldots, n$. Suppose that we are required to recover the values of $f(x)$ for some intermediate values of x, or to obtain the derivative of $f(x)$ for some x in the range $x_0 \le x \le x_n$. The methods for the solution to these problems are based on the concept of the 'differences' of a function which we now proceed to define.

3.3.1 Forward Differences

If $y_0, y_1, y_2, \ldots, y_n$ denote a set of values of y, then $y_1 - y_0, y_2 - y_1, \ldots, y_n - y_{n-1}$ are called the *differences* of y. Denoting these differences by $\Delta y_0, \Delta y_1, \ldots, \Delta y_{n-1}$ respectively, we have

$$\Delta y_0 = y_1 - y_0, \qquad \Delta y_1 = y_2 - y_1, \ \ldots, \qquad \Delta y_{n-1} = y_n - y_{n-1},$$

where Δ is called the *forward difference operator* and $\Delta y_0, \Delta y_1, \ldots$, are called *first forward differences*. The differences of the first forward differences are called *second forward differences* and are denoted by $\Delta^2 y_0, \Delta^2 y_1, \ldots$ Similarly, one can define *third forward differences*, *fourth forward differences*, etc.

Thus,

$$\begin{aligned}\Delta^2 y_0 = \Delta y_1 - \Delta y_0 &= y_2 - y_1 - (y_1 - y_0)\\ &= y_2 - 2y_1 + y_0,\\ \Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0 &= y_3 - 2y_2 + y_1 - (y_2 - 2y_1 + y_0)\\ &= y_3 - 3y_2 + 3y_1 - y_0\\ \Delta^4 y_0 = \Delta^3 y_1 - \Delta^3 y_0 &= y_4 - 3y_3 + 3y_2 - y_1 - (y_3 - 3y_2 + 3y_1 - y_0)\\ &= y_4 - 4y_3 + 6y_2 - 4y_1 + y_0.\end{aligned}$$

It is, therefore, clear that any higher-order difference can easily be expressed in terms of the ordinates, since the coefficients occurring on the right side are the binomial coefficients.

Table 3.1 shows how the forward differences of all orders can be formed:

Table 3.1 Forward Difference Table

x	y_0	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
x_0	y_0						
		Δy_0					
x_1	y_1		$\Delta^2 y_0$				
		Δy_1		$\Delta^3 y_0$			
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$		
		Δy_2		$\Delta^3 y_1$		$\Delta^5 y_0$	
x_3	y_3		$\Delta^2 y_2$		$\Delta^4 y_1$		$\Delta^6 y_0$
		Δy_3		$\Delta^3 y_2$		$\Delta^5 y_1$	
x_4	y_4		$\Delta^2 y_3$		$\Delta^4 y_2$		
		Δy_4		$\Delta^3 y_3$			
x_5	y_5		$\Delta^2 y_4$				
		Δy_5					
x_6	y_6						

In practical computations, the forward difference table can be formed in the following way. For the data points (x_i, y_i), $i = 0, 1, 2, \ldots, n$ and $x_i = x_0 + ih$, we have

$$\Delta y_j = y_{j+1} - y_i, j = 0, 1, \ldots, n - 1.$$

Denoting y_j as DEL(0, j), the above equation can be written as

$$\Delta y_j = \text{DEL}(0, j + 1) - \text{DEL}(0, j) = \text{DEL}(1, j)$$

It follows that

$$\Delta^i y_j = \text{DEL}(i - 1, j + 1) - \text{DEL}(i - 1, j),$$

which is the ith forward difference of y_j.

For the data points (x_i, y_i), $i = 0, 1, 2, \ldots, 6$, we have difference Table 3.2.

Table 3.2 Forward Difference Table

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
x_0	DEL(0, 0)						
		DEL(1, 0)					
x_1	DEL(0, 1)		DEL(2, 0)				
		DEL(1, 1)		DEL(3, 0)			
x_2	DEL(0, 2)		DEL(2, 1)		DEL(4, 0)		
		DEL(1, 2)		DEL(3, 1)		DEL(5, 0)	
x_3	DEL(0, 3)		DEL(2, 2)		DEL(4, 1)		DEL(6, 0)
		DEL(1, 3)		DEL(3, 2)		DEL(5, 1)	
x_4	DEL(0, 4)		DEL(2, 3)		DEL(4, 2)		
		DEL(1, 4)		DEL(3, 3)			
x_5	DEL(0, 5)		DEL(2, 4)				
		DEL(1, 5)					
x_6	DEL(0, 6)						

In Table 3.2

$$\begin{aligned}
\text{DEL}(4, 0) &= \text{DEL}(3, 1) - \text{DEL}(3, 0) \\
&= \text{DEL}(2, 2) - \text{DEL}(2, 1) - [\text{DEL}(2, 1) - \text{DEL}(2, 0)] \\
&= \text{DEL}(1, 3) - \text{DEL}(1, 2) - 2[\text{DEL}(1, 2) - \text{DEL}(1, 1)] \\
&\quad + \text{DEL}(1, 1) - \text{DEL}(1, 0) \\
&= \text{DEL}(0, 4) - \text{DEL}(0, 3) - 3[\text{DEL}(0, 3) - \text{DEL}(0, 2)] \\
&\quad + 3[\text{DEL}(0, 2) - \text{DEL}(0, 1)] - [\text{DEL}(0, 1) - \text{DEL}(0, 0)] \\
&= \text{DEL}(0, 4) - 4\text{DEL}(0, 3) + 6\text{DEL}(0, 2) - 4\text{DEL}(0, 1) + \text{DEL}(0, 0) \\
&= y_4 - 4y_3 + 6y_2 - 4y_1 + y_0
\end{aligned}$$

The forward difference table can now be formed by the simple statements:

```
Do i = 1(1)n
Do j = 0(1)n - i
DEL(i, j) = DEL(i - 1, j + 1) - DEL(i - 1, j)
Next j
Next i
End
```

3.3.2 Backward Differences

The differences $y_1 - y_0$, $y_2 - y_1$, …, $y_n - y_{n-1}$ are called first *backward differences* if they are denoted by ∇y_1, ∇y_2, …, ∇y_n respectively, so that

$$\begin{aligned}
\nabla y_1 &= y_1 - y_0, \quad \nabla y_2 = y_2 - y_1, \\
&\vdots \quad \vdots \quad \vdots \quad \vdots \\
\nabla y_n &= y_n - y_{n-1},
\end{aligned}$$

where ∇ is called the *backward difference operator.* In a similar way, one can define backward differences of higher orders.

Thus, we obtain

$$\nabla^2 y_2 = \nabla y_2 - \nabla y_1$$
$$= y_2 - y_1 - (y_1 - y_0) = y_2 - 2y_1 + y_0,$$
$$\nabla^3 y_3 = \nabla^2 y_3 - \nabla^2 y_2$$
$$= y_3 - 3y_2 + 3y_1 - y_0, \text{ etc.}$$

With the same values of x and y as in Table 3.1, a backward difference Table 3.3 can be formed:

Table 3.3 Backward Difference Table

x	y	∇	∇^2	∇^3	∇^4	∇^5	∇^6
x_0	y_0						
x_1	y_1	∇y_1					
x_2	y_2	∇y_2	$\nabla^2 y_2$				
x_3	y_3	∇y_3	$\nabla^2 y_3$	$\nabla^3 y_3$			
x_4	y_4	∇y_4	$\nabla^2 y_4$	$\nabla^3 y_4$	$\nabla^4 y_4$		
x_5	y_5	∇y_5	$\nabla^2 y_5$	$\nabla^3 y_5$	$\nabla^4 y_5$	$\nabla^5 y_5$	
x_6	y_6	∇y_6	$\nabla^2 y_6$	$\nabla^3 y_6$	$\nabla^4 y_6$	$\nabla^5 y_6$	$\nabla^6 y_6$

3.3.3 Central Differences

The *central difference operator* δ is defined by the relations

$$y_1 - y_0 = \delta y_{1/2}, \quad y_2 - y_1 = \delta y_{3/2}, \ldots, \quad y_n - y_{n-1} = \delta y_{n-1/2}.$$

Similarly, higher-order central differences can be defined. With the values of x and y as in the preceding two tables, a central difference Table 3.4 can be formed:

Table 3.4 Central Difference Table

x	y	δ	δ^2	δ^3	δ^4	δ^5	δ^6
x_0	y_0						
		$\delta y_{1/2}$					
x_1	y_1		$\delta^2 y_1$				
		$\delta y_{3/2}$		$\delta^3 y_{3/2}$			
x_2	y_2		$\delta^2 y_2$		$\delta^4 y_2$		
		$\delta y_{5/2}$		$\delta^3 y_{5/2}$		$\delta^5 y_{5/2}$	
x_3	y_3		$\delta^2 y_3$		$\delta^4 y_3$		$\delta^6 y_3$
		$\delta y_{7/2}$		$\delta^3 y_{7/2}$		$\delta^5 y_{7/2}$	
x_4	y_4		$\delta^2 y_4$		$\delta^4 y_4$		
		$\delta y_{9/2}$		$\delta^3 y_{9/2}$			
x_5	y_5		$\delta^2 y_5$				
		$\delta y_{11/2}$					
x_6	y_6						

It is clear from all the four tables that in a definite numerical case, the same numbers occur in the same positions whether we use forward, backward or central differences. Thus, we obtain

$$\Delta y_0 = \nabla y_1 = \delta y_{1/2}, \qquad \Delta^3 y_2 = \nabla^3 y_5 = \delta^3 y_{7/2}, \ldots$$

3.3.4 Symbolic Relations and Separation of Symbols

Difference formulae can easily be established by symbolic methods, using the *shift* operator E and the *averaging* or the *mean* operator μ, in addition to the operators, Δ, ∇ and δ already defined.

The averaging operator μ is defined by the equation:

$$\mu y_r = \frac{1}{2}(y_{r+1/2} + y_{r-1/2}).$$

The shift operator E is defined by the equation:

$$Ey_r = y_{r+1},$$

which shows that the effect of E is to shift the functional value y_r to the next higher value y_{r+1}. A second equation with E gives

$$E^2 y_r = E(Ey_r) = Ey_{r+1} = y_{r+2},$$

and in general,

$$E^n y_r = y_{r+n}.$$

It is now easy to derive a relationship between Δ and E, for we have

$$\Delta y_0 = y_1 - y_0 = Ey_0 - y_0 = (E-1)y_0$$

and hence

$$\Delta \equiv E - 1 \quad \text{or} \quad E \equiv 1 + \Delta. \tag{3.8a}*$$

We can now express any higher-order forward difference in terms of the given function values. For example,

$$\Delta^3 y_0 = (E-1)^3 y_0 = (E^3 - 3E^2 + 3E - 1)y_0 = y_3 - 3y_2 + 3y_1 - y_0.$$

From the definitions, the following relations can easily be established:

$$\left.\begin{aligned} &\nabla = 1 - E^{-1}, \\ &\delta = E^{1/2} - E^{-1/2}, \\ &\mu = (1/2)(E^{1/2} + E^{-1/2}),\ \mu^2 = 1 + (1/4)\delta^2 \\ &\Delta = \nabla E = \delta E^{1/2}. \end{aligned}\right\} \tag{3.8b}$$

*The student should note that Eq. (3.8a) does not mean that the operators E and Δ have any existence as separate entities; it merely implies that the effect of the operator E on y_0 is the same as that of the operator $(1 + \Delta)$ on y_0.

As an example, we prove the relation

$$\mu^2 \equiv 1 + (1/4)\,\delta^2.$$

We have, by definition,

$$\mu y_r = \frac{1}{2}(y_{r+1/2} + y_{r-1/2})$$

$$= \frac{1}{2}(E^{1/2} y_r + E^{-1/2} y_r)$$

$$= \frac{1}{2}(E^{1/2} + E^{-1/2}) y_r.$$

Hence

$$\mu = \frac{1}{2}(E^{1/2} + E^{-1/2})$$

and

$$\mu^2 = \frac{1}{4}(E^{1/2} + E^{-1/2})^2$$

$$= \frac{1}{4}(E + E^{-1} + 2)$$

$$= \frac{1}{4}[(E^{1/2} - E^{-1/2})^2 + 4]$$

$$= \frac{1}{4}(\delta^2 + 4).$$

We therefore have

$$\mu \equiv \sqrt{1 + \frac{1}{4}\delta^2}.$$

Finally, we define the operator D such that

$$Dy(x) = \frac{d}{dx} y(x).$$

To relate D to E, we start with the Taylor's series

$$y(x+h) = y(x) + hy'(x) + \frac{h^2}{2!} y''(x) + \frac{h^3}{3!} y'''(x) + \cdots$$

This can be written in the symbolic form

$$Ey(x) = \left(1 + hD + \frac{h^2 D^2}{2!} + \frac{h^3 D^3}{3!} + \cdots\right) y(x).$$

Since the series in the brackets is the expansion of e^{hD}, we obtain the interesting result

$$E \equiv e^{hD}. \qquad (3.8c)$$

Using the relation (3.8a), a number of useful identities can be derived. This relation is used to separate the effect of E into that of the powers of Δ and this method of separation is called the *method of separation of symbols*. The following examples demonstrate the use of this method.

Example 3.1 Using the method of separation of symbols, show that

$$\Delta^n u_{x-n} = u_x - nu_{x-1} + \frac{n(n-1)}{2}u_{x-2} + \cdots + (-1)^n u_{x-n}.$$

To prove this result, we start with the right-hand side. Thus,

$$\begin{aligned}
&u_x - nu_{x-1} + \frac{n(n-1)}{2}u_{x-2} + \cdots + (-1)^n u_{x-n} \\
&= u_x - nE^{-1}u_x + \frac{n(n-1)}{2}E^{-2}u_x + \cdots + (-1)^n E^{-n}u_x \\
&= \left[1 - nE^{-1} + \frac{n(n-1)}{2}E^{-2} + \cdots + (-1)^n E^{-n}\right]u_x \\
&= (1 - E^{-1})^n u_x \\
&= \left(1 - \frac{1}{E}\right)^n u_x \\
&= \left(\frac{E-1}{E}\right)^n u_x \\
&= \frac{\Delta^n}{E^n}u_x \\
&= \Delta^n E^{-n} u_x \\
&= \Delta^n u_{x-n},
\end{aligned}$$

which is the left-hand side.

Example 3.2 Show that

$$e^x\left(u_0 + x\Delta u_0 + \frac{x^2}{2!}\Delta^2 u_0 + \cdots\right) = u_0 + u_1 x + u_2\frac{x^2}{2!} + \cdots$$

Now,

$$\begin{aligned}
e^x\left(u_0 + x\,\Delta u_0 + \frac{x^2}{2!}\Delta^2 u_0 + \cdots\right) &= e^x\left(1 + x\Delta + \frac{x^2\Delta^2}{2!} + \cdots\right)u_0 \\
&= e^x e^{x\Delta}u_0 = e^{x(1+\Delta)}u_0 \\
&= e^{xE}u_0
\end{aligned}$$

$$= \left(1 + xE + \frac{x^2E^2}{2!} + \cdots\right) u_0$$

$$= u_0 + xu_1 + \frac{x^2}{2!} u_2 + \cdots,$$

which is the required result.

3.4 DETECTION OF ERRORS BY USE OF DIFFERENCE TABLES

Difference tables can be used to check errors in tabular values. Suppose that there is an error of +1 unit in a certain tabular value. As higher differences are formed, the error spreads out fanwise, and is at the same time, considerably magnified, as shown in Table 3.5.

Table 3.5 Detection of Errors using Difference Table

y	Δ	Δ^2	Δ^3	Δ^4	Δ^5
0					
	0				
0		0			
	0		0		
0		0		0	
	0		0		1
0		0		1	
	0		1		–5
0		1		–4	
	1		–3		10
1		–2		6	
	–1		3		–10
0		1		–4	
	0		–1		5
0		0		–1	
	0		0		–1
0		0		0	
	0		0		
0		0			
	0				
0					

This table shows the following characteristics:

(i) The effect of the error increases with the order of the differences.

(ii) The errors in any one column are the binomial coefficients with alternating signs.

(iii) The algebraic sum of the errors in any difference column is zero, and

(iv) The maximum error occurs opposite the function value containing the error. These facts can be used to detect errors by difference tables. We illustrate this by means of an example.

Example 3.3 Consider the following difference table:

x	y	Δ	Δ^2	Δ^3	Δ^4
1	3010				
		414			
2	3424		–36		
		378		–39	
3	3802		–75		+178
		303		+139	
4	4105		+ 64		–271
		367		–132	
5	4472		– 68		+181
		299		+ 49	
6	4771		–19		– 46
		280		+3	
7	5051		–16		
		264			
8	5315				

The term –271 in the fourth difference column has fluctuations of 449 and 452 on either side of it. Comparison with Table 3.5 suggests that there is an error of –45 in the entry for $x = 4$. The correct value of y is therefore 4105 + 45 = 4150, which shows that the last-two digits have been transposed, a very common form of error. The reader is advised to form a new difference table with this correction, and to check that the third differences are now practically constant.

If an error is present in a given data, the differences of some order will become alternating in sign. Hence, higher-order differences should be formed till the error is revealed as in the above example. If there are errors in several tabular values, then it is not easy to detect the errors by differencing.

3.5 DIFFERENCES OF A POLYNOMIAL

Let $y(x)$ be a polynomial of the nth degree so that

$$y(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n.$$

Then we obtain

$$y(x+h) - y(x) = a_0[(x+h)^n - x^n] + a_1[(x+h)^{n-1} - x^{n-1}] + \cdots$$

$$= a_0(nh)\,x^{n-1} + a_1'x^{n-2} + \cdots + a_n',$$

where $a_1', a_2', \ldots, a_n'$ are the new coefficients.

The above equation can be written as

$$\Delta y(x) = a_0(nh)\,x^{n-1} + a_1'x^{n-2} + \cdots + a_n',$$

which shows that the first difference of a polynomial of the nth degree is a polynomial of degree $(n-1)$. Similarly, the second difference will be a polynomial of degree $(n-2)$, and the coefficient of x^{n-2} will be $a_0n(n-1)h^2$.

Thus the nth difference is $a_0 n! h^n$, which is a constant. Hence, the $(n+1)$th, and higher differences of a polynomial of nth degree will be zero. Conversely, if the nth differences of a tabulated function are constant and the $(n+1)$th, $(n+2)$th, …, differences all vanish, then the tabulated function represents a polynomial of degree n. It should be noted that these results hold good only if the values of x are equally spaced. The converse is important in numerical analysis since it enables us to approximate a function by a polynomial if its differences of some order become nearly constant.

3.6 NEWTON'S FORMULAE FOR INTERPOLATION

Given the set of $(n+1)$ values, viz., (x_0, y_0), (x_1, y_1), (x_2, y_2), …, (x_n, y_n), of x and y, it is required to find $y_n(x)$, a polynomial of the nth degree such that y and $y_n(x)$ agree at the tabulated points. Let the values of x be equidistant, i.e. let

$$x_i = x_0 + ih, \quad i = 0, 1, 2, \ldots, n.$$

Since $y_n(x)$ is a polynomial of the nth degree, it may be written as

$$\left.\begin{aligned} y_n(x) &= a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) \\ &\quad + a_3(x - x_0)(x - x_1)(x - x_2) + \cdots \\ &\quad + a_n(x - x_0)(x - x_1)(x - x_2)\ldots(x - x_{n-1}). \end{aligned}\right\} \tag{3.9}$$

Imposing now the condition that y and $y_n(x)$ should agree at the set of tabulated points, we obtain

$$a_0 = y_0;\ a_1 = \frac{y_1 - y_0}{x_1 - x_0} = \frac{\Delta y_0}{h};\ a_2 = \frac{\Delta^2 y_0}{h^2 2!};\ a_3 = \frac{\Delta^3 y_0}{h^3 3!};\ \cdots;\ a_n = \frac{\Delta^n y_0}{h^n n!};$$

Setting $x = x_0 + ph$ and substituting for $a_0, a_1, \ldots, a_n$, Eq. (3.9) gives

$$y_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

$$+ \frac{p(p-1)(p-2)\ldots(p-n+1)}{n!}\Delta^n y_0, \tag{3.10}$$

which is *Newton's forward difference interpolation formula* and is useful for interpolation *near the beginning* of a set of tabular values.

To find the error committed in replacing the function $y(x)$ by means of the polynomial $y_n(x)$, we use Eq. (3.7) to obtain

$$y(x) - y_n(x) = \frac{(x - x_0)(x - x_1)\ldots(x - x_n)}{(n+1)!} y^{(n+1)}(\xi), \qquad x_0 < \xi < x_n \tag{3.11}$$

As remarked earlier we do not have any information concerning $y^{(n+1)}(x)$, and therefore, formula given in Eq. (3.11) is useless in practice. Nevertheless,

if $y^{(n+1)}(x)$ does not vary too rapidly in the interval, a useful estimate of the derivative can be obtained in the following way. Expanding $y(x+h)$ by Taylor's series (see Theorem 1.4), we obtain

$$y(x+h) = y(x) + hy'(x) + \frac{h^2}{2!}y''(x) + \cdots$$

Neglecting the terms containing h^2 and higher powers of h, this gives

$$y'(x) \approx \frac{1}{h}[y(x+h) - y(x)] = \frac{1}{h}\Delta y(x).$$

Writing $y'(x)$ as $Dy(x)$ where $D \equiv d/dx$, the differentiation operator, the above equation gives the operator relation

$$D \equiv \frac{1}{h}\Delta \quad \text{and so} \quad D^{n+1} \equiv \frac{1}{h^{n+1}}\Delta^{n+1}.$$

We thus obtain

$$y^{(n+1)}(x) \approx \frac{1}{h^{n+1}}\Delta^{n+1}y(x). \tag{3.12}$$

Equation (3.11) can, therefore, be written as

$$y(x) - y_n(x) = \frac{p(p-1)(p-2)\ldots(p-n)}{(n+1)!}\Delta^{n+1}y(\xi) \tag{3.13}$$

in which form it is suitable for computation.

Instead of assuming $y_n(x)$ as in Eq. (3.9), if we choose it in the form

$$\begin{aligned} y_n(x) = a_0 &+ a_1(x - x_n) + a_2(x - x_n)(x - x_{n-1}) \\ &+ a_3(x - x_n)(x - x_{n-1})(x - x_{n-2}) + \cdots \\ &+ a_n(x - x_n)(x - x_{n-1})\ldots(x - x_1). \end{aligned}$$

and then impose the condition that y and $y_n(x)$ should agree at the tabulated points x_n, x_{n-1}, …, x_2, x_1, x_0, we obtain (after some simplification)

$$y_n(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \cdots + \frac{p(p+1)\ldots(p+n-1)}{n!}\nabla^n y_n, \tag{3.14}$$

where $p = (x - x_n)/h$.

This is *Newton's backward difference interpolation formula* and it uses tabular values to the left of y_n. This formula is therefore useful for interpolation *near the end of* the tabular values.

It can be shown that the error in this formula may be written as

$$y(x) - y_n(x) = \frac{p(p+1)(p+2)\ldots(p+n)}{(n+1)!}\nabla^{n+1}y(\xi), \tag{3.15}$$

where $x_0 < x < x_n$ and $x = x_n + ph$.

The following examples illustrate the use of these formulae.

Example 3.4 Find the cubic polynomial which takes the following values: $y(1) = 24$, $y(3) = 120$, $y(5) = 336$, and $y(7) = 720$. Hence, or otherwise, obtain the value of $y(8)$.

We form the difference table:

x	y	Δ	Δ^2	Δ^3
1	24			
		96		
3	120		120	
		216		48
5	336		168	
		384		
7	720			

Here $h = 2$. With $x_0 = 1$, we have $x = 1 + 2p$ or $p = (x - 1)/2$. Substituting this value of p in Eq. (3.10), we obtain

$$y(x) = 24 + \frac{x-1}{2}(96) + \frac{\left(\frac{x-1}{2}\right)\left(\frac{x-1}{2}-1\right)}{2}(120) + \frac{\left(\frac{x-1}{2}\right)\left(\frac{x-1}{2}-1\right)\left(\frac{x-1}{2}-2\right)}{6}(48)$$

$$= x^3 + 6x^2 + 11x + 6.$$

To determine $y(8)$, we observe that $p = 7/2$. Hence, Eq. (3.10) gives:

$$y(8) = 24 + \frac{7}{2}(96) + \frac{(7/2)(7/2-1)}{2}(120) + \frac{(7/2)(7/2-1)(7/2-2)}{6}(48) = 990.$$

Direct substitution in $y(x)$ also yields the same value.

Note: This process of finding the value of y for some value of x *outside* the given range is called extrapolation and this example demonstrates the fact that if a tabulated function is a polynomial, then both interpolation and extrapolation would give exact values.

Example 3.5 Using Newton's forward difference formula, find the sum

$$S_n = 1^3 + 2^3 + 3^3 + \cdots + n^3.$$

We have

$$S_{n+1} = 1^3 + 2^3 + 3^3 + \cdots + n^3 + (n+1)^3$$

Hence

$$S_{n+1} - S_n = (n+1)^3,$$

or

$$\Delta S_n = (n+1)^3. \tag{i}$$

It follows that

$$\Delta^2 S_n = \Delta S_{n+1} - \Delta S_n = (n+2)^3 - (n+1)^3 = 3n^2 + 9n + 7,$$

$$\Delta^3 S_n = 3(n+1)^2 + 9n + 7 - (3n^2 + 9n + 7) = 6n + 12$$

$$\Delta^4 S_n = 6(n+1) + 12 - (6n + 12) = 6.$$

Since $\Delta^5 S_n = \Delta^6 S_n = \cdots = 0$, S_n is a fourth-degree polynomial in n.

Further,

$$S_1 = 1, \qquad \Delta S_1 = 8, \qquad \Delta^2 S_1 = 19, \qquad \Delta^3 S_1 = 18, \qquad \Delta^4 S_1 = 6.$$

Equation (3.10) gives

$$S_n = 1 + (n-1)(8) + \frac{(n-1)(n-2)}{2}(19) + \frac{(n-1)(n-2)(n-3)}{6}(18)$$

$$+ \frac{(n-1)(n-2)(n-3)(n-4)}{24}(6)$$

$$= \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2$$

$$= \left[\frac{n(n+1)}{2}\right]^2.$$

Example 3.6 Values of x (in degrees) and sin x are given in the following table:

x (in degrees)	sin x
15	0.2588190
20	0.3420201
25	0.4226183
30	0.5
35	0.5735764
40	0.6427876

Determine the value of sin 38°.

The difference table is

x	sin x	Δ	Δ^2	Δ^3	Δ^4	Δ^5
15	0.2588190					
		0.0832011				
20	0.3420201		–0.0026029			
		0.0805982		–0.0006136		
25	0.4226183		–0.0032165		0.0000248	
		0.0773817		–0.0005888		0.0000041
30	0.5		–0.0038053		0.0000289	
		0.0735764		–0.0005599		
35	0.5735764		–0.0043652			
		0.0692112				
40	0.6427876					

To find sin 38°, we use Newton's backward difference formula with $x_n = 40$ and $x = 38$. This gives

$$p = \frac{x - x_n}{h} = \frac{38 - 40}{5} = -\frac{2}{5} = -0.4.$$

Hence, using Eq. (3.14), we obtain

$$\begin{aligned} y(38) &= 0.6427876 - 0.4\,(0.0692112) + \frac{-0.4(-0.4+1)}{2}(-0.0043652) \\ &\quad + \frac{(-0.4)(-0.4+1)(-0.4+2)}{6}(-0.0005599) \\ &\quad + \frac{(-0.4)(-0.4+1)(-0.4+2)(-0.4+3)}{24}(0.0000289) \\ &\quad + \frac{(-0.4)(-0.4+1)(-0.4+2)(-0.4+3)(-0.4+4)}{120}(0.0000041) \\ &= 0.6427876 - 0.02768448 + 0.00052382 + 0.00003583 - 0.00000120 \\ &= 0.6156614. \end{aligned}$$

Example 3.7 Find the missing term in the following table:

x	y
0	1
1	3
2	9
3	–
4	81

Explain why the result differs from $3^3 = 27$.

Since four points are given, the given data can be approximated by a third degree polynomial in x. Hence $\Delta^4 y_0 = 0$. Substituting $\Delta = E - 1$ and simplifying, we get

$$E^4 y_0 - 4E^3 y_0 + 6E^2 y_0 - 4Ey_0 + y_0 = 0.$$

Since $E^r y_0 = y_r$, the above equation becomes

$$y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0.$$

Substituting for y_0, y_1, y_2 and y_4 in the above, we obtain

$$y_3 = 31.$$

The tabulated function is 3^x and the exact value of $y(3)$ is 27. The error is due to the fact that the exponential function 3^x is approximated by means of a polynomial in x of degree 3.

Example 3.8 The table below gives the values of tan x for $0.10 \le x \le 0.30$:

x	$y = \tan x$
0.10	0.1003
0.15	0.1511
0.20	0.2027
0.25	0.2553
0.30	0.3093

Find : (a) tan 0.12 (b) tan 0.26, (c) tan 0.40 and (d) tan 0.50.

The table of difference is

x	y	Δ	Δ^2	Δ^3	Δ^4
0.10	0.1003				
		0.0508			
0.15	0.1511		0.0008		
		0.0516		0.0002	
0.20	0.2027		0.0010		0.0002
		0.0526		0.0004	
0.25	0.2553		0.0014		
		0.0540			
0.30	0.3093				

(a) To find tan (0.12), we have $0.12 = 0.10 + p(0.05)$, which gives $p = 0.4$. Hence, Eq. (3.10) gives

$$\tan(0.12) = 0.1003 + 0.4(0.0508) + \frac{0.4(0.4-1)}{2}(0.0008) + \frac{0.4(0.4-1)(0.4-2)}{6}(0.0002) + \frac{0.4(0.4-1)(0.4-2)(0.4-3)}{24}(0.0002)$$

$$= 0.1205.$$

(b) To find tan (0.26), we have $0.26 = 0.30 + p(0.05)$, which gives $p = -0.8$. Hence, Eq. (3.14) gives

$$\tan(0.26) = 0.3093 - 0.8(0.0540) + \frac{-0.8(-0.8+1)}{2}(0.0014) + \frac{-0.8(-0.8+1)(-0.8+2)}{6}(0.0004) + \frac{-0.8(-0.8+1)(-0.8+2)(-0.8+3)}{24}(0.0002) = 0.2662.$$

Proceeding as in the case (i) above, we obtain

(c) tan (0.40) = 0.4241, and

(d) tan (0.50) = 0.5543.

The actual values, correct to four decimal places, of tan (0.12), tan (0.26), tan (0.40) and tan (0.50) are respectively 0.1206, 0.2660, 0.4228 and 0.5463. Comparison of the computed and actual values shows that in the first-two cases (i.e. of interpolation) the results obtained are fairly accurate whereas in the last-two cases (i.e. of extrapolation) the errors are quite considerable. The example therefore demonstrates the important result that if a tabulated function is other than a polynomial, then extrapolation very far from the table limits would be dangerous—although interpolation can be carried out very accurately.

3.7 CENTRAL DIFFERENCE INTERPOLATION FORMULAE

In the preceding section, we derived and discussed Newton's forward and backward interpolation formulae, which are applicable for interpolation near the beginning and end respectively, of tabulated values. We shall, in the present section, discuss the central difference formulae which are most suited for interpolation near the middle of a tabulated set. The central difference operator δ was already introduced in Section 3.3.3.

The most important central difference formulae are those due to Stirling, Bessel and Everett. These will be discussed in Sections 3.7.2, 3.7.3 and 3.7.4, respectively. Gauss's formulae, introduced in Section 3.7.1 below, are of interest from a theoretical stand-point only.

3.7.1 Gauss' Central Difference Formulae

In this section, we will discuss Gauss' forward and backward formulae.

Gauss' forward formula

We consider the following difference table in which the central ordinate is taken for convenience as y_0 corresponding to $x = x_0$.

The differences used in this formula lie on the line shown in Table 3.6. The formula is, therefore, of the form

$$y_p = y_0 + G_1\Delta y_0 + G_2\Delta^2 y_{-1} + G_3\Delta^3 y_{-1} + G_4\Delta^4 y_{-2} + \cdots \qquad (3.16)$$

where G_1, G_2, … have to be determined. The y_p on the left side can be expressed in terms of y_0, Δy_0 and higher-order differences of y_0, as follows:

Table 3.6 Gauss' Forward Formula

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
x_{-3}	y_{-3}						
		Δy_{-3}					
x_{-2}	y_{-2}		$\Delta^2 y_{-3}$				
		Δy_{-2}		$\Delta^3 y_{-3}$			
x_{-1}	y_{-1}		$\Delta^2 y_{-2}$		$\Delta^4 y_{-3}$		
		Δy_{-1}		$\Delta^3 y_{-2}$		$\Delta^5 y_{-3}$	
x_0	y_0		$\Delta^2 y_{-1}$		$\Delta^4 y_{-2}$		$\Delta^6 y_{-3}$
		Δy_0		$\Delta^3 y_{-1}$		$\Delta^5 y_{-2}$	
x_1	y_1		$\Delta^2 y_0$		$\Delta^4 y_{-1}$		
		Δy_1		$\Delta^3 y_0$			
x_2	y_2		$\Delta^2 y_1$				
		Δy_2					
x_3	y_3						

Clearly,

$$y_p = E^p y_0$$

$$= (1+\Delta)^p y_0, \text{ using relation (3.8a)}$$

$$= y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

Similarly, the right side of Eq. (3.16) can also be expressed in terms of y_0, Δy_0 and higher-order differences. We have

$$\Delta^2 y_{-1} = \Delta^2 E^{-1} y_0$$

$$= \Delta^2 (1+\Delta)^{-1} y_0$$

$$= \Delta^2 (1 - \Delta + \Delta^2 - \Delta^3 + \cdots) y_0$$

$$= \Delta^2 (y_0 - \Delta y_0 + \Delta^2 y_0 - \Delta^3 y_0 + \cdots)$$

$$= \Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 + \cdots$$

$$\Delta^3 y_{-1} = \Delta^3 y_0 - \Delta^4 y_0 + \Delta^5 y_0 - \Delta^6 y_0 + \cdots$$

$$\Delta^4 y_{-2} = \Delta^4 E^{-2} y_0$$

$$= \Delta^4 (1+\Delta)^{-2} y_0$$

$$= \Delta^4 (y_0 - 2\Delta y_0 + 3\Delta^2 y_0 - 4\Delta^3 y_0 + \cdots)$$

$$= \Delta^4 y_0 - 2\Delta^5 y_0 + 3\Delta^6 y_0 - 4\Delta^7 y_0 + \cdots$$

Hence Eq. (3.16) gives the identity

$$y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0 + \cdots$$

$$= y_0 + G_1 \Delta y_0 + G_2(\Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 + \cdots) + G_3(\Delta^3 y_0 - \Delta^4 y_0 + \Delta^5 y_0 - \Delta^6 y_0 + \cdots) + G_4(\Delta^4 y_0 - 2\Delta^5 y_0 + 3\Delta^6 y_0 - 4\Delta^7 y_0 + \cdots) + \cdots \quad (3.17)$$

Equating the coefficients of Δy_0, $\Delta^2 y_0$, $\Delta^3 y_0$, etc., on both sides of Eq. (3.17), we obtain

$$\left.\begin{aligned} G_1 &= p, \\ G_2 &= \frac{p(p-1)}{2!}, \\ G_3 &= \frac{(p+1)\,p(p-1)}{3!}, \\ G_4 &= \frac{(p+1)\,p(p-1)(p-2)}{4!}. \end{aligned}\right\} \quad (3.18)$$

Gauss' backward formula

This formula uses the differences which lie on the line shown in Table 3.7.

Table 3.7 Gauss' Backward Formula

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
⋮	⋮						
x_{-1}	y_{-1}						
		Δy_{-1}		$\Delta^3 y_{-2}$		$\Delta^5 y_{-3}$	
x_0	y_0		$\Delta^2 y_{-1}$		$\Delta^4 y_{-2}$		$\Delta^6 y_{-3}$
		Δy_0		$\Delta^3 y_{-1}$		$\Delta^5 y_{-2}$	
x_1	y_1						
⋮	⋮						

Gauss' backward formula can therefore be assumed to be of the form

$$y_p = y_0 + G_1'\Delta y_{-1} + G_2'\Delta^2 y_{-1} + G_3'\Delta^3 y_{-2} + G_4'\Delta^4 y_{-2} + \cdots \tag{3.19}$$

where G_1', $G_2', \ldots$ have to be determined. Following the same procedure as in Gauss' forward formula, we obtain

$$\left.\begin{aligned} G_1' &= p, \\ G_2' &= \frac{p(p+1)}{2!}, \\ G_3' &= \frac{(p+1)\,p(p-1)}{3!} \\ G_4' &= \frac{(p+2)(p+1)\,p(p-1)}{4!} \\ &\vdots \end{aligned}\right\} \tag{3.20}$$

Example 3.9 From the following table, find the value of $e^{1.17}$ using Gauss' forward formula:

x	e^x
1.00	2.7183
1.05	2.8577
1.10	3.0042
1.15	3.1582
1.20	3.3201
1.25	3.4903
1.30	3.6693

We have

$$1.17 = 1.15 + p(0.05),$$

which gives

$$p = \frac{0.02}{0.05} = \frac{1}{4}.$$

The difference table is given below.

x	e^x	Δ	Δ^2	Δ^3	Δ^4
1.00	2.7183				
		0.1394			
1.05	2.8577		0.0071		
		0.1465		0.0004	
1.10	3.0042		0.0075		0
		0.1540		0.0004	
1.15	3.1582		0.0079		0
		0.1619		0.0004	
1.20	3.3201		0.0083		0.0001
		0.1702		0.0005	
1.25	3.4903		0.0088		
		0.1790			
1.30	3.6693				

Using formulae (3.16) and (3.18), we obtain

$$e^{1.17} = 3.1582 + \frac{2}{5}(0.1619) + \frac{(2/5)(2/5-1)}{2}(0.0079)$$

$$+ \frac{(2/5+1)\,(2/5)\,(2/5-1)}{6}(0.0004)$$

$$= 3.1582 + 0.0648 - 0.0009$$

$$= 3.2221.$$

3.7.2 Stirling's Formula

Taking the mean of Gauss' forward and backward formulae, we obtain

$$y_p = y_0 + p\frac{\Delta y_{-1} + \Delta y_0}{2} + \frac{p^2}{2}\Delta^2 y_{-1} + \frac{p(p^2-1)}{3!}\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}$$

$$+ \frac{p^2(p^2-1)}{4!}\Delta^4 y_{-2} + \cdots \qquad (3.21)$$

Formula given in Eq. (3.21) is called *Stirling's formula*.

3.7.3 Bessel's Formula

This is a very useful formula for practical interpolation, and it uses the differences as shown in the following table, where the brackets mean that the average of the values has to be taken.

⋮	⋮										
x_{-1}	y_{-1}										
x_0	$\begin{pmatrix} y_0 \\ y_1 \end{pmatrix}$	Δy_0	$\begin{pmatrix} \Delta^2 y_{-1} \\ \Delta^2 y_0 \end{pmatrix}$	$\Delta^3 y_{-1}$	$\begin{pmatrix} \Delta^4 y_{-2} \\ \Delta^4 y_{-1} \end{pmatrix}$	$\Delta^5 y_{-2}$	$\begin{pmatrix} \Delta^6 y_{-3} \\ \Delta^6 y_{-2} \end{pmatrix}$				
x_1											
⋮	⋮										

Hence, Bessel's formula can be assumed in the form

$$y_p = \frac{y_0 + y_1}{2} + B_1 \Delta y_0 + B_2 \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + B_3 \Delta^3 y_{-1} + B_4 \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \cdots$$

$$= y_0 + \left(B_1 + \frac{1}{2} \right) \Delta y_0 + B_2 \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + B_3 \Delta^3 y_{-1} + B_4 \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \cdots \tag{3.22}$$

Using the method outlined in Section 3.7.1, i.e., Gauss' forward formula, we obtain

$$\left.\begin{aligned} B_1 + \frac{1}{2} &= p, \\ B_2 &= \frac{p(p-1)}{2!}, \\ B_3 &= \frac{p(p-1)(p-1/2)}{3!}, \\ B_4 &= \frac{(p+1)\, p(p-1)(p-1)}{4!}, \\ &\vdots \end{aligned}\right\} \tag{3.23}$$

Hence, Bessel's interpolation formula may be written as

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{p(p-1)(p-1/2)}{3!} \Delta^3 y_{-1} + \frac{(p+1)\, p(p-1)(p-2)}{4!} \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \cdots \tag{3.24}$$

3.7.4 Everett's Formula

This is an extensively used interpolation formula and uses only even order differences, as shown in the following table:

x_0	y_0		$\Delta^2 y_{-1}$		$\Delta^4 y_{-2}$		$\Delta^6 y_{-3}$
		–		–		–	
x_1	y_1		$\Delta^2 y_0$		$\Delta^4 y_{-1}$		$\Delta^6 y_{-2}$

Hence the formula has the form

$$y_p = E_0 y_0 + E_2 \Delta^2 y_{-1} + E_4 \Delta^4 y_{-2} + \cdots + F_0 y_1 + F_2 \Delta^2 y_0 + F_4 \Delta^4 y_{-1} + \cdots \quad (3.25)$$

The coefficients $E_0, F_0, E_2, F_2, E_4, F_4, \ldots$ can be determined by the same method as in the preceding cases, and we obtain

$$\left.\begin{aligned} E_0 &= 1 - p = q, & F_0 &= p, \\ E_2 &= \frac{q(q^2 - 1^2)}{3!}, & F_2 &= \frac{p(p^2 - 1^2)}{3!}, \\ E_4 &= \frac{q(q^2 - 1^2)(q^2 - 2^2)}{5!}, & F_4 &= \frac{p(p^2 - 1^2)(p^2 - 2^2)}{5!}, \\ &\vdots & &\vdots \end{aligned}\right\} \quad (3.26)$$

Hence Everett's formula is given by

$$\left.\begin{aligned} y_p = q y_0 + \frac{q(q^2 - 1^2)}{3!}\Delta^2 y_{-1} + \frac{q(q^2 - 1^2)(q^2 - 2^2)}{5!}\Delta^4 y_{-2} + \cdots \\ + p y_1 + \frac{p(p^2 - 1^2)}{3!}\Delta^2 y_0 + \frac{p(p^2 - 1^2)(p^2 - 2^2)}{5!}\Delta^4 y_{-1} + \cdots \end{aligned}\right\} \quad (3.27)$$

where $q = 1 - p$.

3.7.5 Relation between Bessel's and Everett's Formulae

These formulae are very closely related, and it is possible to deduce one from the other by a suitable rearrangement. To see this we start with Bessel's formula

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{p(p-1)(p-1/2)}{3!}\Delta^3 y_{-1}$$

$$+ \frac{(p+1)\,p(p-1)(p-2)}{4!}\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \cdots$$

$$= y_0 + p(y_1 - y_0) + \frac{p(p-1)}{2!}\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{p(p-1)(p-1/2)}{3!}(\Delta^2 y_0 - \Delta^2 y_{-1})$$
$$+ \frac{(p+1)\,p(p-1)(p-2)}{4!}\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \cdots$$

expressing the odd order differences in terms of low even order differences. This gives on simplification

$$y_p = (1-p)y_0 + \left[\frac{p(p-1)}{4} - \frac{(p-1)\,p(p-1/2)}{6}\right]\Delta^2 y_{-1} + \cdots$$
$$+ py_1 + \left[\frac{p(p-1)}{4} + \frac{p(p-1)(p-1/2)}{6}\right]\Delta^2 y_0 + \cdots$$
$$= qy_0 + \frac{q(q^2-1^2)}{3!}\Delta^2 y_{-1} + \cdots + py_1 + \frac{p(p^2-1^2)}{3!}\Delta^2 y_0 + \cdots$$

which is *Everett's formula* truncated after second differences. Hence we have a result of practical importance that Everett's formula truncated after second differences is equivalent to Bessel's formula truncated after third differences. In a similar way, Bessel's formula may be deduced from Everett's.

3.8 PRACTICAL INTERPOLATION

In the preceding sections, we have derived some interpolation formulae of great practical importance. A natural question is: Which one of these formulae gives the most accurate result?

(i) For interpolation at the beginning or end of a table of values, Newton's forward and backward interpolation formulae have to be used respectively.

(ii) For interpolation near the middle of a set of values, the following are the choices:

$$\text{Stirling's formula if } -\frac{1}{4} \le p \le \frac{1}{4},$$

and

$$\text{Bessel's formula for } \frac{1}{4} \le p \le \frac{3}{4}.$$

It can be shown that if the third differences are negligible, then Bessel's formula is about seven times more accurate than Stirling's formula. If the third differences are more than 60 in magnitude, then Everett's formula should be preferred.

Example 3.10 The following table gives the values of e^x for certain equidistant values of x. Find the value of e^x when $x = 0.644$.

x	$y = e^x$
0.61	1.840431
0.62	1.858928
0.63	1.877610
0.64	1.896481
0.65	1.915541
0.66	1.934792
0.67	1.954237

The table of differences is

x	$y = e^x$	Δ	Δ^2	Δ^3	Δ^4
0.61	1.840431				
		0.018497			
0.62	1.858928		0.000185		
		0.018682		0.000004	
0.63	1.877610		0.000189		–0.000004
		0.018871		0	
0.64	1.896481		0.000189		0.000002
		0.019060		0.000002	
0.65	1.915541		0.000191		0.000001
		0.019251		0.000003	
0.66	1.934792		0.000194		
		0.019445			
0.67	1.954237				

Clearly,

$$p = \frac{0.644 - 0.64}{0.01} = 0.4.$$

The third difference contribution to both Stirling's and Bessel's formulae is negligible, and using Stirling's formula, we obtain

$$y(0.644) = 1.896481 + 0.4\frac{0.018871 + 0.019060}{2} + \frac{0.16}{2}(0.000189)$$

$$= 1.896481 + 0.0075862 + 0.00001512$$

$$= 1.904082,$$

while Bessel's formula gives

$$y(0.644) = 1.896481 + 0.4(0.019060) + \frac{0.4(0.4-1)}{2} \cdot \frac{0.000189 + 0.000191}{2}$$

$$= 1.896481 + 0.0076240 - 0.0000228$$

$$= 1.904082.$$

Using Everett's formula, we find that

$$y(0.644) = 0.6(1.896481) + \frac{0.6(0.36-1)}{2}(0.000189)$$
$$+ 0.4(1.915541) + \frac{0.4(0.16-1)}{2}(0.000191)$$
$$= 1.1378886 - 0.000012096 + 0.7662164 - 0.000010696$$
$$= 1.904082.$$

In all the above cases, the value obtained is correct to six decimal places.

It is known from algebra that the nth degree polynomial which passes through $(n+1)$ points is *unique*. Hence the various interpolation formulae derived here are actually only different forms of the same polynomial. It, therefore, follows that all the interpolation formulae should give the same functional value. This is illustrated in the above example where we found that the interpolated value of 0.644 is 1.904082 regardless of which formula is used.

Example 3.11 From the table of Example 3.10, find the value of e^x when $x = 0.638$, using Stirling's and Bessel's formulae.

It was mentioned in Section 3.8 that Stirling's formula gives the most accurate result for $-1/4 \le p \le 1/4$, and Bessel's formula is most efficient for $1/4 \le p \le 3/4$. In order to use these formulae, we therefore, have to choose x_0 so that p satisfies the appropriate inequality.

To use Stirling's formula, we choose $x_0 = 0.64$ and $x_n = 0.638$ so that $p = -0.2$. Hence,

$$y(0.638) = 1.896481 - 0.2 \cdot \frac{0.018871 + 0.019060}{2} + \frac{0.04}{2}(0.000189)$$
$$= 1.896481 - 0.0037931 + 0.0000038$$
$$= 1.892692,$$

which is correct to the last decimal place.

For Bessel's formula, we choose $x_0 = 0.63$, $x_n = 0.638$ so that $p = 0.8$. Hence, we obtain

$$y(0.638) = 1.877610 + 0.8(0.018871) + \frac{0.8(0.8-1)}{2}(0.000189)$$
$$= 1.877610 + 0.0150968 - 0.0000151$$
$$= 1.892692, \text{ as before.}$$

Example 3.12 The values of x and e^{-x} are given in the following table. Find the value of e^{-x} when $x = 1.7475$.

x	$y = e^{-x}$	Δ	Δ^2	Δ^3	Δ^4
1.72	0.1790661479				
		–17817379			
1.73	0.1772844100		177285		
		–17640094		–1762	
1.74	0.1755204006		175523		13
		–17464571		–1749	
1.75	0.1737739435		173774		22
		–17290797		–1727	
1.76	0.1720448638		172047		15
		–17118750		–1712	
1.77	0.1703329888		170335		
		–16948415			
1.78	0.1686381473				

It should be noted that in writing the differences in the above table, the zeros between the decimal point and the first significant digit to its right are omitted. Thus, in the column of second differences, the number 173774 should be taken as 0.0000173774 in the computations.

To compute $y(1.7475)$, we choose $x_0 = 1.74$ and $x_p = 1.7475$ so that $p = 3/4$. We shall obtain the solution by using both Bessel's and Everett's formulae.

(i) If we use Bessel's formula, the third differences need to be taken into account since they exceed 60 units in magnitude. Hence Bessel's formula gives

$$y(1.7475) = 0.1755204006 - \frac{3}{4}(0.0017464571) + \frac{(3/4)(3/4-1)}{2}\frac{0.0000175523 + 0.0000173774}{2}$$

$$= 0.1755204006 - 0.00130984284 - 0.00000163734 + 0.00000000137$$

$$= 0.1742089218, \text{ correct to ten decimal places.}$$

(ii) On the other hand, if we use Everett's formula up to second differences only, we obtain

$$y(1.7475) = \frac{1}{4}(0.1755204006) + \frac{(1/4)(1/16-1)}{6}(0.0000175523) + \frac{3}{4}(0.1737739435) + \frac{(3/4)(9/16-1)}{6}(0.0000173774)$$

$$= 0.04388010015 - 0.00000068564 + 0.13033045764 - 0.00000095033$$

$$= 0.1742089218, \text{ as before.}$$

This example verifies the result of Section 3.7.5 that Everett's formula turncated after second differences is equivalent to Bessel's formula truncated after third differences. When the fourth difference contribution becomes significant (i.e. when they exceed 20 units in magnitude), Everett's formula will be easier to apply since it uses only the even order differences.

3.9 INTERPOLATION WITH UNEVENLY SPACED POINTS

In the preceding sections, we have derived interpolation formulae of utmost importance and discussed their practical use in some detail. But, as is well known, they possess the disadvantage of requiring the values of the independent variable to be equally spaced. It is therefore desirable to have interpolation formulae with unequally spaced values of the argument. We discuss, in the present section and the next, four such formulae: (i) Lagrange's interpolation formula which uses only the function values, (ii) Hermite's interpolation formula which is similar to Lagrange's formula, (iii) Newton's general interpolation formula which uses what are called divided differences and (iv) Aitken's method of interpolation by iteration.

3.9.1 Lagrange's Interpolation Formula

Let $y(x)$ be continuous and differentiable $(n + 1)$ times in the interval (a, b). Given the $(n + 1)$ points (x_0, y_0), (x_1, y_1), …, (x_n, y_n) where the values of x need not necessarily be equally spaced, we wish to find a polynomial of degree n, say $L_n(x)$, such that

$$L_n(x_i) = y(x_i) = y_i, \quad i = 0, 1, \ldots, n \tag{3.28}$$

Before deriving the general formula, we first consider a simpler case, viz., the equation of a straight line (a linear polynomial) passing through two points (x_0, y_0) and (x_1, y_1). Such a polynomial, say $L_1(x)$, is easily seen to be

$$\begin{aligned} L_1(x) &= \frac{x - x_1}{x_0 - x_1} y_0 + \frac{x - x_0}{x_1 - x_0} y_1 \\ &= l_0(x) y_0 + l_1(x) y_1 \\ &= \sum_{i=0}^{1} l_i(x) y_i, \end{aligned} \tag{3.29}$$

where

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} \quad \text{and} \quad l_1(x) = \frac{x - x_0}{x_1 - x_0}. \tag{3.30}$$

From Eq. (3.30), it is seen that

$$l_0(x_0) = 1, \qquad l_0(x_1) = 0, \qquad l_1(x_0) = 0, \qquad l_1(x_1) = 1.$$

These relations can be expressed in a more convenient form as

$$l_i(x_j) = \begin{cases} 1, \text{ if } i = j \\ 0, \text{ if } i \neq j. \end{cases} \tag{3.31}$$

The $l_i(x)$ in Eq. (3.29) also have the property

$$\sum_{i=0}^{1} l_i(x) = l_0(x) + l_1(x) = \frac{x - x_1}{x_0 - x_1} + \frac{x - x_0}{x_1 - x_0} = 1. \tag{3.32}$$

Equation (3.29) is the *Lagrange polynomial of degree one passing through* two points (x_0, y_0) and (x_1, y_1). In a similar way, the *Lagrange polynomial of degree two* passing through three points (x_0, y_0), (x_1, y_1) and (x_2, y_2) is written as

$$\begin{aligned} L_2(x) &= \sum_{i=0}^{2} l_i(x)\, y_i \\ &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2, \end{aligned} \tag{3.33}$$

where the $l_i(x)$ satisfy the conditions given in Eqs. (3.31) and (3.32).

To derive the general formula, let

$$L_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \tag{3.34}$$

be the desired polynomial of the nth degree such that conditions given in Eq. (3.28) (called the *interpolatory conditions*) are satisfied. Substituting these conditions in Eq. (3.34), we obtain the system of equations

$$\left.\begin{aligned} y_0 &= a_0 + a_1 x_0 + a_2 x_0^2 + \cdots + a_n x_0^n \\ y_1 &= a_0 + a_1 x_1 + a_2 x_1^2 + \cdots + a_n x_1^n \\ y_2 &= a_0 + a_1 x_2 + a_2 x_2^2 + \cdots + a_n x_2^n \\ &\vdots \\ y_n &= a_0 + a_1 x_n + a_2 x_n^2 + \cdots + a_n x_n^n. \end{aligned}\right\} \tag{3.35}$$

The set of Eqs. (3.35) will have a solution if

$$\begin{vmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{vmatrix} \neq 0. \tag{3.36}$$

The value of this determinant, called *Vandermonde's determinant*, is

$$(x_0 - x_1)(x_0 - x_2)\ldots(x_0 - x_n)(x_1 - x_2)\ldots(x_1 - x_n)\ldots(x_{n-1} - x_n).$$

Eliminating a_0, a_1, …, a_n from Eqs. (3.34) and (3.35), we obtain

$$\begin{vmatrix} L_n(x) & 1 & x & x^2 & \cdots & x^n \\ y_0 & 1 & x_0 & x_0^2 & \cdots & x_0^n \\ y_1 & 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ y_n & 1 & x_n & x_n^2 & \cdots & x_n^n \end{vmatrix} = 0, \tag{3.37}$$

which shows that $L_n(x)$ is a linear combination of y_0, y_1, y_2, …, y_n. Hence we write

$$L_n(x) = \sum_{i=0}^{n} l_i(x) y_i, \tag{3.38}$$

where $l_i(x)$ are polynomials in x of degree n. Since $L_n(x_j) = y_j$ for $j = 0, 1, 2, \ldots, n$, Eq. (3.32) gives

$$\left.\begin{aligned} l_i(x_j) &= 0 \qquad \text{if} \quad i \neq j \\ l_j(x_j) &= 1 \qquad \text{for all} \quad j \end{aligned}\right\},$$

which are the same as Eq. (3.31). Hence $l_i(x)$ may be written as

$$l_i(x) = \frac{(x - x_0)(x - x_1)\ldots(x - x_{i-1})(x - x_{i+1})\ldots(x - x_n)}{(x_i - x_0)(x_i - x_1)\ldots(x_i - x_{i-1})(x_i - x_{i+1})\ldots(x_i - x_n)}, \tag{3.39}$$

which obviously satisfies the conditions (3.31).

If we now set

$$\Pi_{n+1}(x) = (x - x_0)(x - x_1)\ldots(x - x_{i-1})(x - x_i)(x - x_{i+1})\ldots(x - x_n), \tag{3.40}$$

then

$$\Pi'_{n+1}(x_i) = \frac{d}{dx}[\Pi_{n+1}(x)]_{x = x_i}$$

$$= (x_i - x_0)(x_i - x_1)\ldots(x_i - x_{i-1})(x_i - x_{i+1})\ldots(x_i - x_n) \tag{3.41}$$

so that Eq. (3.39) becomes

$$l_i(x) = \frac{\Pi_{n+1}(x)}{(x - x_i)\,\Pi'_{n+1}(x_i)}. \tag{3.42}$$

Hence Eq. (3.38) gives

$$L_n(x) = \sum_{i=0}^{n} \frac{\Pi_{n+1}(x)}{(x - x_i)\,\Pi'_{n+1}(x_i)}\, y_i, \tag{3.43}$$

which is called *Lagrange's interpolation formula*. The coefficients $l_i(x)$, defined in Eq. (3.39), are called *Lagrange interpolation coefficients*. Interchanging x and y in Eq. (3.43), we obtain the formula

$$L_n(y) = \sum_{i=0}^{n} \frac{\Pi_{n+1}(y)}{(y - y_i)\,\Pi'_{n+1}(y_i)}\, x_i, \tag{3.44}$$

which is useful for *inverse interpolation*.

It is trivial to show that the Lagrange interpolating polynomial is *unique*. To prove this, we assume the contrary. Let $\bar{L}_n(x)$ be a polynomial, distinct from $L_n(x)$, of degree not exceeding n and such that

$$\bar{L}_n(x_i) = y_i, \qquad i = 0,\ 1,\ 2,\ \ldots,\ n.$$

Then the polynomial defined by $M(x)$, where

$$M(x) = L_n(x) - \bar{L}_n(x)$$

vanishes at the $(n + 1)$ points x_i, $i = 0, 1, \ldots, n$. Hence we have

$$M_n(x) \equiv 0,$$

which shows that $L_n(x)$ and $\bar{L}_n(x)$ are identical.

A major advantage of this formula is that the coefficients in Eq. (3.44) are easily determined. Further, it is more general in that it is applicable to either equal or unequal intervals and the abscissae $x_0, x_1, \ldots, x_n$ need not be in order. Using this formula it is, however, inconvenient to pass from one interpolation polynomial to another of degree one greater.

The following examples illustrate the use of Lagrange's formula.

Example 3.13 Certain corresponding values of x and $\log_{10} x$ are (300, 2.4771), (304, 2.4829), (305, 2.4843) and (307, 2.4871). Find $\log_{10}301$.

From formula given in Eq. (3.43), we obtain

$$\begin{aligned}
\log_{10} 301 &= \frac{(-3)\,(-4)\,(-6)}{(-4)\,(-5)\,(-7)}(2.4771) + \frac{(1)\,(-4)\,(-6)}{(4)\,(-1)\,(-3)}(2.4829) \\
&\quad + \frac{(1)\,(-3)\,(-6)}{(5)\,(1)\,(-2)}(2.4843) + \frac{(1)\,(-3)\,(-4)}{(7)\,(3)\,(2)}(2.4871) \\
&= 1.2739 + 4.9658 - 4.4717 + 0.7106 \\
&= 2.4786.
\end{aligned}$$

Example 3.14 If $y_1 = 4$, $y_3 = 12$, $y_4 = 19$ and $y_x = 7$, find x.

Using Eq. (3.44), we have

$$x = \frac{(-5)(-12)}{(-8)(-15)}(1) + \frac{(3)(-12)}{(8)(-7)}(3) + \frac{(3)(-5)}{(15)(7)}(4)$$

$$= \frac{1}{2} + \frac{27}{14} - \frac{4}{7}$$

$$= 1.86.$$

The actual value is 2.0 since the above values were obtained from the polynomial $y(x) = x^2 + 3$.

Example 3.15 Find the Lagrange interpolating polynomial of degree 2 approximating the function $y = \ln x$ defined by the following table of values. Hence determine the value of ln 2.7.

x	$y = \ln x$
2	0.69315
2.5	0.91629
3.0	1.09861

We have

$$l_0(x) = \frac{(x-2.5)(x-3.0)}{(-0.5)(-1.0)} = 2x^2 - 11x + 15.$$

Similarly, we find

$$l_1(x) = -(4x^2 - 20x + 24) \quad \text{and} \quad l_2(x) = 2x^2 - 9x + 10.$$

Hence

$$L_2(x) = (2x^2 - 11x + 15)(0.69315) - (4x^2 - 20x + 24)(0.91629)$$
$$+ (2x^2 - 9x + 10)(1.09861)$$
$$= -0.08164x^2 + 0.81366x - 0.60761,$$

which is the required quadratic polynomial.

Putting $x = 2.7$, in the above polynomial, we obtain

$$\ln 2.7 \approx L_2(2.7) = -0.08164(2.7)^2 + 0.81366(2.7) - 0.60761 = 0.9941164.$$

Actual value of ln 2.7 = 0.9932518, so that

$$|\text{Error}| = 0.0008646.$$

Example 3.16 The function $y = \sin x$ is tabulated below

x	$y = \sin x$
0	0
$\pi/4$	0.70711
$\pi/2$	1.0

Using Lagrange's interpolation formula, find the value of $\sin(\pi/6)$.

We have

$$\sin\frac{\pi}{6} \approx \frac{(\pi/6-0)(\pi/6-\pi/2)}{(\pi/4-0)(\pi/4-\pi/2)}(0.70711) + \frac{(\pi/6-0)(\pi/6-\pi/4)}{(\pi/2-0)(\pi/2-\pi/4)}(1)$$

$$= \frac{8}{9}(0.70711) - \frac{1}{9}$$

$$= \frac{4.65688}{9}$$

$$= 0.51743.$$

Example 3.17 Using Lagrange's interpolation formula, find the form of the function $y(x)$ from the following table

x	y
0	-12
1	0
3	12
4	24

Since $y = 0$ when $x = 1$, it follows that $x - 1$ is a factor. Let $y(x) = (x-1)\,R(x)$. Then $R(x) = y/(x-1)$. We now tabulate the values of x and $R(x)$.

x	$R(x)$
0	12
3	6
4	8

Applying Lagrange's formula to the above table, we find

$$R(x) = \frac{(x-3)(x-4)}{(-3)(-4)}(12) + \frac{(x-0)(x-4)}{(3-0)(3-4)}(6) + \frac{(x-0)(x-3)}{(4-0)(4-3)}(8)$$

$$= (x-3)(x-4) - 2x(x-4) + 2x(x-3)$$

$$= x^2 - 5x + 12.$$

Hence the required polynomial approximation to $y(x)$ is given by

$$y(x) = (x-1)(x^2 - 5x + 12).$$

3.9.2 Error in Lagrange's Interpolation Formula

Equation (3.7) can be used to estimate the error of the Lagrange interpolation formula for the class of functions which have continuous derivatives of order up to $(n+1)$ on $[a, b]$. We, therefore, have

$$y(x) - L_n(x) = R_n(x) = \frac{\Pi_{n+1}(x)}{(n+1)!} y^{(n+1)}(\xi), \quad a < \xi < b \tag{3.45}$$

and the quantity E_L, where

$$E_L = \max_{[a,b]} |R_n(x)| \tag{3.46}$$

may be taken as an estimate of error. Further, if we assume that

$$|y^{(n+1)}(\xi)| \le M_{n+1}, \quad a \le \xi \le b \tag{3.47}$$

then

$$E_L \le \frac{M_{n+1}}{(n+1)!} \max_{[a,b]} |\Pi_{n+1}(x)| \tag{3.48}$$

The following examples illustrate the computation of the error.

Example 3.18 Estimate the error in the value of y obtained in Example 3.15.

Since $y = \ln x$, we obtain $y' = 1/x$, $y'' = -1/x^2$ and $y''' = 2/x^3$. It follows that $y'''(\xi) = 2/\xi^3$. Thus the continuity conditions on $y(x)$ and its derivatives are satisfied in [2, 3]. Hence

$$R_n(x) = \frac{(x-2)(x-2.5)(x-3)}{6} \frac{2}{\xi^3}, \qquad 2 < \xi < 3$$

But

$$\left| \frac{1}{\xi^3} \right| < \frac{1}{2^3} = \frac{1}{8}.$$

When $x = 2.7$, we therefore obtain

$$|R_n(x)| \le \left| \frac{(2.7-2)(2.7-2.5)(2.7-3)}{6} \frac{2}{8} \right| = \frac{0.7 \times 0.2 \times 0.3}{3 \times 8} = 0.00175,$$

which agrees with the actual error given in Example 3.15.

Example 3.19 Estimate the error in the solution computed in Example 3.16.

Since $y(x) = \sin x$, we have

$$y'(x) = \cos x, \qquad y''(x) = -\sin x, \qquad y'''(x) = -\cos x.$$

Hence $|y'''(\xi)| < 1$.

When $x = \pi/6$.

$$|R_n(x)| \le \left| \frac{(\pi/6 - 0)(\pi/6 - \pi/4)(\pi/6 - \pi/2)}{6} \right| = \frac{1}{6}\frac{\pi}{6}\frac{\pi}{12}\frac{\pi}{3} = 0.02392,$$

which agrees with the actual error in the solution obtained in Example 3.16.

3.9.3 Hermite's Interpolation Formula

The interpolation formulae so far considered make use of only a certain number of function values. We now derive an interpolation formula in which both the function and its first derivative values are to be assigned at each point of interpolation. This is referred to as *Hermite's interpolation formula*. The interpolation problem is then defined as follows: Given the set of data points (x_i, y_i, y_i'), $i = 0, 1, \ldots, n$, it is required to determine a polynomial of the least degree, say $H_{2n+1}(x)$, such that

$$H_{2n+1}(x_i) = y_i \quad \text{and} \quad H'_{2n+1}(x_i) = y_i'; \qquad i = 0, 1, \ldots, n, \tag{3.49}$$

where the primes denote differentiation with respect to x. The polynomial $H_{2n+1}(x)$ is called *Hermite's interpolation polynomial*. We have here $(2n + 2)$ conditions and therefore the number of coefficients to be determined is $(2n + 2)$ and the degree of the polynomial is $(2n + 1)$. In analogy with the Lagrange interpolation formula (3.43), we seek a representation of the form

$$H_{2n+1}(x) = \sum_{i=0}^{n} u_i(x) y_i + \sum_{i=0}^{n} v_i(x) y_i', \tag{3.50}$$

where $u_i(x)$ and $v_i(x)$ are polynomials in x of degree $(2n + 1)$. Using conditions (3.49), we obtain

$$\left.\begin{aligned} u_i(x_j) &= \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \ne j \end{cases}; \quad v_i(x) = 0, \text{ for all } i \\ u_i'(x) &= 0, \text{ for all } i; \qquad v_i'(x_j) = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \ne j. \end{cases} \end{aligned}\right\} \tag{3.51}$$

Since $u_i(x)$ and $v_i(x)$ are polynomials in x of degree $(2n + 1)$, we write

$$u_i(x) = A_i(x)\,[l_i(x)]^2 \quad \text{and} \quad v_i(x) = B_i(x)\,[l_i(x)]^2, \tag{3.52}$$

where $l_i(x)$ are given by Eq. (3.42). It is easy to see that $A_i(x)$ and $B_i(x)$ are both linear functions in x. We therefore write

$$u_i(x) = (a_i x + b_i)\,[l_i(x)]^2 \quad \text{and} \quad v_i(x) = (c_i x + d_i)\,[l_i(x)]^2 \tag{3.53}$$

Using conditions Eq. (3.51) in Eq. (3.53), we obtain

$$\left.\begin{aligned} a_i x_i + b_i &= 1 \\ c_i x_i + d_i &= 0 \end{aligned}\right\} \tag{3.54a}$$

and

$$\left.\begin{aligned} a_i + 2l_i'(x_i) &= 0 \\ c_i &= 1. \end{aligned}\right\} \tag{3.54b}$$

From Eq. (3.54), we deduce

$$\left.\begin{aligned} a_i &= -2l_i'(x_i), \quad b_i = 1 + 2x_i\ l_i'(x_i) \\ c_i &= 1, \qquad\qquad d_i = -x_i. \end{aligned}\right\} \tag{3.55}$$

Hence Eq. (3.53) become

$$u_i(x) = [-2x\ l_i'(x_i) + 1 + 2x_i l_i'(x_i)][l_i(x)]^2$$

$$= [1 - 2\ (x - x_i)\ l_i'(x_i)][l_i(x)]^2 \tag{3.56a}$$

and

$$v_i(x) = (x - x_i)\,[l_i(x)]^2. \tag{3.56b}$$

Using the above expressions for $u_i(x)$ and $v_i(x)$ in Eq. (3.50), we obtain finally

$$H_{2n+1}(x) = \sum_{i=0}^{n} [1 - 2(x - x_i)\ l_i'(x_i)][l_i(x)]^2 y_i + \sum_{i=0}^{n} (x - x_i)\,[l_i(x)]^2 y_i', \tag{3.57}$$

which is the required *Hermite interpolation formula*.

The following examples demonstrate the application of Hermite's formula.

Example 3.20 Find the third-order Hermite polynomial passing through the points $(x_i,\ y_i,\ y'_i)$, $i = 0,\ 1$.

Putting $n = 1$ in Hermite's formula (3.57), we obtain

$$H_3(x) = [1 - 2(x - x_0)\ l_0'(x_0)][l_0(x)]^2 y_0 + [1 - 2\ (x - x_1)\ l_1'(x_1)][l_1(x)]^2 y_1$$
$$+ (x - x_0)\,[l_0(x)]^2\ y_0' + (x - x_1)\,[l_1(x)]^2 y_1'. \tag{i}$$

Since

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x_1 - x}{h_1} \quad \text{and} \quad l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - x_0}{h_1},$$

where $h_1 = x_1 - x_0$. Hence

$$l_0'(x) = -\frac{1}{h_1} \quad \text{and} \quad l_1'(x) = \frac{1}{h_1}.$$

Then, Eq. (i) simplifies to

$$H_3(x)=\left[1+\frac{2(x-x_0)}{h_1}\right]\frac{(x_1-x)^2}{h_1^2}y_0+\left[1+\frac{2(x_1-x)}{h_1}\right]\frac{(x-x_0)^2}{h_1^2}y_1$$

$$+(x-x_0)\frac{(x_1-x)^2}{h_1^2}y_0'+(x-x_1)\frac{(x-x_0)^2}{h_1^2}y_1', \qquad \text{(ii)}$$

which is the required Hermite formula.

Example 3.21 Determine the Hermite polynomial of degree 5, which fits the following data and hence find an approximate value of ln 2.7.

x	$y=\ln x$	$y'=1/x$
2.0	0.69315	0.5
2.5	0.91629	0.4000
3.0	1.09861	0.33333

The polynomials $l_i(x)$ have already been computed in Example 3.15. These are

$$l_0(x)=2x^2-11x+15, \qquad l_1(x)=-(4x^2-20x+24), \qquad l_2(x)=2x^2-9x+10.$$

We therefore obtain

$$l_0'(x)=4x-11, \qquad l_1'(x)=-8x+20, \qquad l_2'(x)=4x-9.$$

Hence

$$l_0'(x_0)=-3, \qquad l_1'(x_1)=0, \qquad l_2'(x_2)=3$$

Equations (3.56) give

$$u_0(x)=(6x-11)\,(2x^2-11x+15)^2, \qquad v_0(x)=(x-2)\,(2x^2-11x+15)^2$$

$$u_1(x)=(4x^2-20x+24)^2, \qquad v_1(x)=(x-2.5)\,(4x^2-20x+24)^2,$$

$$u_2(x)=(19-6x)\,(2x^2-9x+10)^2, \qquad v_2(x)=(x-3)\,(2x^2-9x+10)^2,$$

Substituting these expressions in Eq. (3.57), we obtain the required Hermite polynomial

$$\begin{aligned}H_5(x)=&(6x-11)\,(2x^2-11x+15)^2\,(0.69315)\\&+(4x^2-20x+24)^2\,(0.91629)\\&+(19-6x)\,(2x^2-9x+10)^2(1.09861)\\&+(x-2)\,(2x^2-11x+15)^2\,(0.5)\\&+(x-2.5)\,(4x^2-20x+24)^2(0.4)\\&+(x-3)\,(2x^2-9x+10)^2\,(0.33333).\end{aligned}$$

Putting $x = 2.7$ and simplifying, we obtain

$$\ln (2.7) \approx H_5(2.7) = 0.993252,$$

which is correct to six decimal places. This is therefore a more accurate result than that obtained by using the Lagrange interpolation formula.

3.10 DIVIDED DIFFERENCES AND THEIR PROPERTIES

The Lagrange interpolation formula, derived in Section 3.9.1, has the disadvantage that if another interpolation point were added, then the interpolation coefficients $l_i(x)$ will have to be recomputed. We therefore seek an interpolation polynomial which has the property that a polynomial of higher degree may be derived from it by simply adding new terms. Newton's general interpolation formula is one such formula and it employs what are called *divided differences*. It is our principal purpose in this section to define such differences and discuss certain of their properties to obtain the basic formula due to Newton.

Let $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$ be the given $(n+1)$ points. Then the divided differences of order $1, 2, \ldots, n$ are defined by the relations:

$$\left.\begin{aligned} [x_0, x_1] &= \frac{y_1 - y_0}{x_1 - x_0}, \\ [x_0, x_1, x_2] &= \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0}, \\ &\vdots \\ [x_0, x_1, \ldots, x_n] &= \frac{[x_1, x_2, \ldots, x_n] - [x_0, x_1, \ldots, x_{n-1}]}{x_n - x_0}. \end{aligned}\right\} \tag{3.58}$$

Even if the arguments are equal, the divided differences may still have a meaning. We then set $x_1 = x_0 + \varepsilon$ so that

$$\begin{aligned} [x_0, x_1] &= \lim_{\varepsilon \to 0} [x_0, x_0 + \varepsilon] \\ &= \lim_{\varepsilon \to 0} \frac{y(x_0 + \varepsilon) - y(x_0)}{\varepsilon} \\ &= y'(x_0), \quad \text{if } y(x) \text{ is differentiable.} \end{aligned}$$

Similarly,

$$\underbrace{[x_0, x_0, \ldots, x_0]}_{(r+1) \text{ arguments}} = \frac{y^r(x_0)}{r!}. \tag{3.59}$$

From Eq. (3.58), it is easy to see that

$$[x_0, x_1] = \frac{y_0}{x_0 - x_1} + \frac{y_1}{x_1 - x_0} = [x_1, x_0].$$

Again,

$$\begin{aligned}[x_0, x_1, x_2] &= \frac{1}{x_2 - x_0}\left(\frac{y_2 - y_1}{x_2 - x_1} - \frac{y_1 - y_0}{x_1 - x_0}\right) \\ &= \frac{1}{x_2 - x_0}\left[\frac{y_2}{x_2 - x_1} - y_1\left(\frac{1}{x_2 - x_1} + \frac{1}{x_1 - x_0}\right) + \frac{y_0}{x_1 - x_0}\right] \\ &= \frac{y_0}{(x_0 - x_1)(x_0 - x_2)} + \frac{y_1}{(x_1 - x_0)(x_1 - x_2)} \\ &\quad + \frac{y_2}{(x_2 - x_0)(x_2 - x_1)}. \end{aligned} \tag{3.60}$$

Similarly it can be shown that

$$\begin{aligned}[x_0, x_1, \ldots, x_n] &= \frac{y_0}{(x_0 - x_1)\ldots(x_0 - x_n)} + \frac{y_1}{(x_1 - x_0)\ldots(x_1 - x_n)} + \cdots \\ &\quad + \frac{y_n}{(x_n - x_0)\ldots(x_n - x_{n-1})}. \end{aligned} \tag{3.61}$$

Hence the divided differences are symmetrical in their arguments.

Now let the arguments be equally spaced so that $x_1 - x_0 = x_2 - x_1 = \cdots = x_n - x_{n-1} = h$. Then we obtain

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{1}{h}\Delta y_0 \tag{3.62}$$

$$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0} = \frac{1}{2h}\left(\frac{\Delta y_1}{h} - \frac{\Delta y_0}{h}\right) = \frac{1}{2h^2}\Delta^2 y_0 = \frac{1}{h^2 2!}\Delta^2 y_0 \tag{3.63}$$

and in general,

$$[x_0, x_1, \ldots, x_n] = \frac{1}{h^n n!}\Delta^n y_0. \tag{3.64}$$

If the tabulated function is a polynomial of nth degree, then $\Delta^n y_0$ would be a constant and hence the nth divided difference would also be a constant.

For the set of values (x_i, y_i), $i = 0, 1, 2, \ldots, n$, divided differences can be generated by the following statements.

```
Define y(x_j) = y_j = DD(0, j), j = 0, 1, 2, …, n
Do i = 1(1)n
Do j = 0(1)(n - i)
DD(i, j) = (DD(i - 1, j + 1) - DD(i - 1, j)) / (X(i + j) - X(j))
Next j
Next i
```

3.10.1 Newton's General Interpolation Formula

By definition, we have

$$[x, x_0] = \frac{y - y_0}{x - x_0},$$

so that

$$y = y_0 + (x - x_0)[x, x_0] \tag{3.65}$$

Again

$$[x, x_0, x_1] = \frac{[x, x_0] - [x_0, x_1]}{x - x_1}$$

which gives

$$[x, x_0] = [x_0, x_1] + (x - x_1)\,[x, x_0, x_1]$$

Substituting this value of $[x, x_0]$ in Eq. (3.65), we obtain

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x, x_0, x_1] \tag{3.66}$$

But

$$[x, x_0, x_1, x_2] = \frac{[x, x_0, x_1] - [x_0, x_1, x_2]}{x - x_2},$$

and so

$$[x, x_0, x_1] = [x_0, x_1, x_2] + (x - x_2)[x, x_0, x_1, x_2] \tag{3.67}$$

Equation (3.66) now gives

$$\begin{aligned} y = y_0 &+ (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] \\ &+ (x - x_0)\,(x - x_1)(x - x_2)[x, x_0, x_1, x_2] \end{aligned} \tag{3.68}$$

Proceeding in this way, we obtain

$$\begin{aligned} y = y_0 &+ (x - x_0)\,[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] \\ &+ (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3] + \cdots \\ &+ (x - x_0)(x - x_1)(x - x_2)\,\cdots\,(x - x_n)[x, x_0, x_1, \ldots, x_n] \end{aligned} \tag{3.69}$$

This formula is called *Newton's general interpolation formula with divided differences,* the last term being the remainder term after $(n + 1)$ terms.

After generating the divided differences, interpolation can be carried out by the following statements.

Let y_k be required corresponding to the value $x = x_k$. Then

```
yk = y0
factor = 1.0
Do i = 0(1)(n - 1)
factor = factor * (xk - xi)
yk = yk + factor * DD (i + 1, 0)
Next i
End
```

Example 3.22 As our first example to illustrate the use of Newton's divided difference formula, we consider the data of Example 3.13.

The divided difference table is

x	$\log_{10} x$		
300	2.4771		
		0.00145	
304	2.4829		0.00001
		0.00140	
305	2.4843		0
		0.00140	
307	2.4871		

Hence Eq. (3.69) gives

$$\log_{10} 301 = 2.4771 + 0.00145 + (-3)(-0.00001) = 2.4786, \text{ as before.}$$

It is clear that the arithmetic in this method is much simpler when compared to that in Lagrange's method.

Example 3.23 Using the following table find $f(x)$ as a polynomial in x.

x	$f(x)$
-1	3
0	-6
3	39
6	822
7	1611

The divided difference table is

x	$f(x)$				
–1	3				
		–9			
0	–6		6		
		15		5	
3	39		41		1
		261		13	
6	822		132		
		789			
7	1611				

Hence Eq. (3.69) gives

$$f(x) = 3+(x+1)(-9)+x(x+1)(6)+x(x+1)(x-3)(5)+x(x+1)(x-3)(x-6)$$
$$= x^4 - 3x^3 + 5x^2 - 6.$$

3.10.2 Interpolation by Iteration

Newton's general interpolation formula may be considered as one of a class of methods which generate successively higher-order interpolation formulae. We now describe another method of this class, due to A.C. Aitken, which has the advantage of being very easily programmed for a digital computer.

Given the $(n + 1)$ points (x_0, y_0), (x_1, y_1), …, (x_n, y_n), where the values of x need not necessarily be equally spaced, then to find the value of y corresponding to any given value of x we proceed iteratively as follows: obtain a first approximation to y by considering the first-two points only; then obtain its second approximation by considering the first-three points, and so on. We denote the different interpolation polynomials by $\Delta(x)$, with suitable subscripts, so that at the first stage of approximation, we have

$$\Delta_{01}(x) = y_0 + (x - x_0)[x_0, x_1] = \frac{1}{x_1 - x_0}\begin{vmatrix} y_0 & x_0 - x \\ y_1 & x_1 - x \end{vmatrix}. \qquad (3.70)$$

Similarly, we can form $\Delta_{02}(x)$, $\Delta_{03}(x)$, …

Next, we form Δ_{012} by considering the first-three points:

$$\Delta_{012}(x) = \frac{1}{x_2 - x_1}\begin{vmatrix} \Delta_{01}(x) & x_1 - x \\ \Delta_{02}(x) & x_2 - x \end{vmatrix}. \qquad (3.71)$$

Similarly, we obtain $\Delta_{013}(x)$, $\Delta_{014}(x)$, etc. At the nth stage of approximation, we obtain

$$\Delta_{012\ldots n}(x) = \frac{1}{x_n - x_{n-1}}\begin{vmatrix} \Delta_{012\ldots\overline{n-1}}(x) & x_{n-1} - x \\ \Delta_{012\ldots\overline{n-2n}}(x) & x_n - x \end{vmatrix}. \qquad (3.72)$$

The computations may conveniently be arranged as in Table 3.8 below:

Table 3.8 Aitken's Scheme

x	y				
x_0	y_0				
		$\Delta_{01}(x)$			
x_1	y_1		$\Delta_{012}(x)$		
		$\Delta_{02}(x)$		$\Delta_{0123}(x)$	
x_2	y_2		$\Delta_{013}(x)$		$\Delta_{01234}(x)$
		$\Delta_{03}(x)$		$\Delta_{0124}(x)$	
x_3	y_3		$\Delta_{014}(x)$		
		$\Delta_{04}(x)$			
x_4	y_4				

Example 3.26 The following table gives the values of z for different values of x and y. Find z when $x = 2.5$ and $y = 1.5$.

	x				
y	0	1	2	3	4
0	0	1	4	9	16
1	2	3	6	11	18
2	6	7	10	15	22
3	12	13	16	21	28
4	18	19	22	27	34

We first interpolate with respect to x keeping y constant. For $x = 2.5$, we obtain the following table using *linear interpolation*.

y	z
0	6.5
1	8.5
2	12.5
3	18.5
4	24.5

Now, we interpolate with respect to y using linear interpolation once again. For $y = 1/5$, we obtain

$$z = \frac{8.5 + 12.5}{2} = 10.5$$

so that $z(2.5, 1.5) = 10.5$. Actually, the tabulated function is $z = x^2 + y^2 + y$ and hence $z(2.5, 1.5) = 10.0$, so that the computed value has an error of 5%.

EXERCISES

3.1 Form a table of differences for the function

$$f(x) = x^3 + 5x - 7$$

for $x = -1, 0, 1, 2, 3, 4, 5$. Continue the table to obtain $f(6)$ and $f(7)$.

3.2 Evaluate

(a) $\Delta^2 x^3$ (b) $\Delta^2(\cos x)$ (c) $\Delta[(x + 1)(x + 2)]$

(d) $\Delta(\tan^{-1} x)$ (e) $\Delta\left[\dfrac{f(x)}{g(x)}\right]$.

3.3 Locate and correct the error in the following table:

x	2.5	3.0	3.5	4.0	4.5	5.0	5.5
y	4.32	4.83	5.27	5.47	6.26	6.79	7.23

3.4 Locate and correct the error in the following table:

x	1.00	1.05	1.10	1.15	1.20	1.25	1.30
e^x	2.7183	2.8577	3.0042	3.1528	3.3201	3.4903	3.6693

3.5 Prove the following:

(a) $y_x = y_{x-1} + \Delta y_{x-2} + \Delta^2 y_{x-3} + \cdots + \Delta^{n-1} y_{x-n} + \Delta^n y_{x-(n+1)}$

(b) $\Delta^n y_x = y_{x+n} - {}^nC_1\, y_{x+n-1} + {}^nC_2 y_{x+n-2} + \cdots + (-1)^n y_x$

(c) $y_1 + y_2 + \cdots + y_n = {}^nC_1 y_1 + {}^nC_2\, \Delta y_1 + \cdots + \Delta^{n-1} y_1$

3.6 From the following table, find the number of students who obtained marks between 60 and 70:

Marks obtained	0–40	40–60	60–80	80–100	100–120
No. of students	250	120	100	70	50

3.7 Find the polynomial which approximates the following values:

x	3	4	5	6	7	8	9
y	13	21	31	43	57	73	91

If the number 31 is the fifth term of the series, find the first and the tenth terms of the series.

3.8 Find $f(0.23)$ and $f(0.29)$ from the following table:

x	0.20	0.22	0.24	0.26	0.28	0.30
$f(x)$	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

3.9 Prove that

(a) $\Delta = \mu\delta + \dfrac{\delta^2}{2}$ (b) $\Delta^3 y_2 = \nabla^3 y_5$

3.10 From the table of cubes given below, find $(6.36)^3$ and $(6.61)^3$.

x	6.1	6.2	6.3	6.4	6.5	6.6	6.7
x^3	226.981	238.328	250.047	262.144	274.625	287.496	300.763

3.11 Define the operators Δ, ∇, δ, E and E^{-1} and show that

(a) $\Delta^r y_k = \nabla^r y_{k+r} = \delta^r y_{k+\frac{r}{2}}$ (b) $\Delta\nabla y_k = \nabla\Delta y_k = \delta^2 y_k$

(c) $\mu\delta = \dfrac{\Delta + \nabla}{2}$ (d) $1 + \mu^2\delta^2 = \left(1 + \dfrac{1}{2}\delta^2\right)^2$

(e) $\Delta^2 = (1 + \Delta)\delta^2$ (f) $\Delta\left(\dfrac{1}{y_k}\right) = -\dfrac{\Delta y_k}{y_k y_{k+1}}$.

3.12 Show that

$$\left(\Delta - \frac{1}{2}\delta^2\right) = \delta\left(1 + \frac{\delta^2}{4}\right)^{1/2}$$

3.13 Find the missing terms in the following:

x	0	5	10	15	20	25	30
y	1	3	?	73	225	?	1153

3.14 Derive expressions for the errors in Newton's formulae of forward and backward differences. Estimate the maximum error made in any value of sin x in Example 3.6 obtained by interpolation in the range $15° \le x \le 40°$.

3.15 Certain values of x and $f(x)$ are given below. Find $f(1.235)$.

x	1.00	1.05	1.10	1.15	1.20	1.25
$f(x)$	0.682689	0.706282	0.728668	0.749856	0.769861	0.788700

3.16 Prove the following relations:

(a) $\delta^2 E = \Delta^2$ (b) $E^{-1/2} = \mu - \frac{\delta}{2}$ (c) $\nabla = \delta E^{-1/2}$

(d) $\Delta - \nabla = \delta^2$ (e) $\mu = \cosh \frac{hD}{2}$.

3.17 Using Gauss's forward formula, find the value of $f(32)$ given that $f(25) = 0.2707$, $f(30) = 0.3027$, $f(35) = 0.3386$ and $f(40) = 0.3794$.

3.18 State Gauss's backward formula and use it to find the value of $\sqrt{12525}$, given that $\sqrt{12500} = 111.8034$, $\sqrt{12510} = 111.8481$, $\sqrt{12520} = 111.8928$, $\sqrt{12530} = 111.9375$ and $\sqrt{12540} = 111.9822$.

3.19 State Stirling's formula for interpolation at the middle of a table of values and find $e^{1.91}$ from the following table:

x	1.7	1.8	1.9	2.0	2.1	2.2
e^x	5.4739	6.0496	6.6859	7.3891	8.1662	9.0250

3.20 Using Stirling's formula, find cos(0.17), given that cos(0) = 1, cos(0.05) = 0.9988, cos(0.10) = 0.9950, cos(0.15) = 0.9888, cos(0.20) = 0.9801, cos(0.25) = 0.9689, and cos(0.30) = 0.9553.

3.21 State Bessel's formula for interpolation and mention its limitations. Use this formula to solve the problem in 3.20.

3.22 The complete elliptic integral of the second kind is defined as

$$k(m) = \int_0^{\pi/2} \frac{1}{\sqrt{1 - m\sin^2\theta}}\, d\theta$$

Find $k(0.25)$ given that
$k(0.20) = 1.6596$, $k(0.22) = 1.6698$, $k(0.24) = 1.6804$, $k(0.26) = 1.6912$, $k(0.28) = 1.7024$, $k(0.30) = 1.7139$.

3.23 Using Bessel's formula, find $y(5)$ given that
$y(0) = 14.27$, $y(4) = 15.81$, $y(8) = 17.72$, and $y(12) = 19.96$.

3.24 From Bessel's formula, derive the following formula for midway interpolation:

$$y_{1/2} = \frac{1}{2}(y_0 + y_1) - \frac{1}{16}\left(\Delta^2 y_{-1} + \Delta^2 y_0\right) + \frac{3}{256}\left(\Delta^4 y_{-2} + \Delta^4 y_{-1}\right) + \cdots$$

Evaluate sin(0.20) given that
sin(0.15) = 0.1494, sin(0.17) = 0.1692, sin(0.19) = 0.1889, sin(0.21) = 0.2085, sin(0.23) = 0.2280.

3.25 Deduce Everett's formula from Bessel's formula and show that Everett's formula truncated after second differences is equivalent to Bessel's formula truncated after third differences. Use Everett's formula to find cos(12.5°) given that
cos(0°) = 1, cos(5°) = 0.9962, cos(10°) = 0.9848, cos(15°) = 0.9659, cos(20°) = 0.9397.

3.26 Using Everett's formula, evaluate $f(25)$ from the set of values
$f(20) = 2854$, $f(24) = 3162$, $f(28) = 3544$, $f(32) = 3992$.

3.27 State Lagrange's interpolation formula and find a bound for the error in linear interpolation.

3.28 Write an algorithm for Lagrange's formula. Find the polynomial which fits the following data

(–1, 7), (1, 5) and (2, 15).

3.29 Find $y(2)$ from the following data using Lagrange's formula

x	0	1	3	4	5
y	0	1	81	256	625

3.30 Let the values of the function $y = \sin x$ be tabulated at the abscissae 0, $\pi/4$ and $\pi/2$. If the Lagrange polynomial $L_2(x)$ is fitted to this data, find a bound for the error in the interpolated value.

3.31 Find a cubic polynomial which fits the data

(–2, –12), (–1, –8), (2, 3) and (3, 5).

3.32 Show that

$$\sum_{i=0}^{n} \frac{\Pi_{n+1}(x)}{(x - x_i)\Pi'_{n+1}(x_i)} = 1,$$

where $\Pi_{n+1}(x) = (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_n)$.

3.33 In complex analysis, the residue theorem is used in the evaluation of contour integrals. If a function $f(z)$ is analytic inside and on a closed contour C, and $x_0, x_1, x_2, \ldots, x_n$ are simple poles inside C, then

$$\int_C f(z)dz = 2\pi i \quad \text{(sum of the residues at } x_0, x_1, \ldots, x_n)$$

[The residue of $f(z)$ at $z = a$ is defined as $\lim_{z \to a} (z - a) f(z)$]

If $x_0, x_1, x_2, \ldots, x_n$ are simple poles of a function $y(x)$ which is analytic inside and an a closed contour C, then show that

$$|y(x) - L_n(x)| = \frac{1}{2\pi i} \int_C \frac{y(t)\Pi_{n+1}(x)\,dt}{(t-x)\Pi_{n+1}(t)}$$

3.34 Lagrange's formula can be used to express a rational function as a sum of partial fractions (see, Stanton [1967]). Express

$$f(x) = \frac{x^2 + x - 3}{x^3 - 2x^2 - x + 2}$$

as a sum of partial fractions.

3.35 Establish Newton's divided-difference interpolation formula and give an estimate of the remainder term. Deduce Newton's forward and backward difference interpolation formulae as particular cases.

3.36 Given

$$f(x) = \frac{1}{x^2},$$

find the divided differences $[a, b]$ and $[a, b, c]$.

3.37 Given the set of tabulated points (0, 2), (1, 3), (2, 12) and (15, 3587) satisfying the function $y = f(x)$, compute $f(4)$ using Newton's divided difference formula.

3.38 The nth divided difference $[x_0, x_1, x_2, \ldots, x_n]$ can be expressed as the quotient of two determinants. Show that

$$[x_0, x_1, x_2] = \frac{\begin{vmatrix} 1 & 1 & 1 \\ x_0 & x_1 & x_2 \\ y_0 & y_1 & y_2 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ x_0 & x_1 & x_2 \\ x_0^2 & x_1^2 & x_2^2 \end{vmatrix}}$$

3.39 Show that

$$[x_0,\ x_1] = \int_0^1 y'\,(x_0\,t_0 + x_1 t_1)\,dt_1,$$

where $t_0 \geq 0$ and $t_0 + t_1 = 1$.

3.40 Tabulate values of $\sin x$ for $x = 0(0.01)$, 1.0 and find the maximum errors in linear and quadratic interpolations using Lagrange's formula.

3.41 If $f(x) = \dfrac{1}{x}$, prove that

$$[x_0, x_1, \ldots, x_r] = \frac{(-1)^r}{x_0\, x_1 \cdots x_r}.$$

3.42 Using Hermite's interpolation formula, estimate the value of ln 4.2 from the data (values of x, ln x and $\dfrac{1}{x}$):
(4.0, 1.38629, 0.25000), (4.5, 1.50408, 0.22222), (5.0, 1.60944, 0.20000).

3.43 Find the Hermite polynomial of the third degree approximating the function $y(x)$ such that

$$y(0) = 1, \quad y'(0) = 0,$$
$$y(1) = 3, \quad y'(1) = 5.$$

3.44 Values of x and $\sqrt[3]{x}$ are given below
(51, 3.708), (55, 3.803), (57, 3.848). Find x when $\sqrt[3]{x} = 3.780$.

3.45 From the table of values of x and e^x, viz. (1.4, 4.0552), (1.5, 4.4817), (1.6, 4.9530), (1.7, 5.4739), find x when $e^x = 4.7115$, using the method of successive approximations.

3.46 The second degree polynomial which satisfies the set of values (0, 1), (1, 2) and (2, 1) is
(a) $1 + 2x - x^2$ (b) $1 - 2x + x^2$ (c) $1 - 2x - x^2$ (d) $1 + 2x + x^2$
Find the correct alternative in the above.

3.47 If $\Delta y = 1 + 2x + 3x^2$, which one of the following is not true?

(a) $\Delta^2 y = 6x + 5$ (b) $\Delta^3 y = 6$

(c) $\Delta^4 y = 0$ (d) $y = x^2 + x^3$

3.48 Which of the following statements are true?

(a) $\Delta(\tan^{-1}x) = \tan^{-1}\dfrac{h}{1+x(x+h)}$ (b) $\Delta(\cos 2x) = 2\sin x\,(x + h)$

(c) $\Delta^2(x^3) = 6x$ (d) $\delta = \nabla E^{1/2}$

(e) $\Delta^2 = (1 - \Delta)\delta^2$ (f) $\mu\delta = \dfrac{\Delta + \nabla}{2}$

(g) $\Delta x^n = nx^{n-1}$ (h) $\left(\frac{\Delta^2}{E}\right)x^3 = 6x$

(i) $\Delta^n e^x = e^x$

3.49 For the function $z = f(x, y)$, the following values are given

$$f(0, 0) = 0, \quad f(0, 1) = 2, \quad f(0, 2) = 6,$$
$$f(1, 0) = 1, \quad f(1, 1) = 3, \quad f(1, 2) = 7,$$
$$f(2, 0) = 4, \quad f(2, 1) = 6, \quad f(2, 2) = 10.$$

Estimate the value of $f(0.5, 0.5)$ by the method of linear interpolation and compare the result with the actual value obtained from $z = x^2 + y^2 + y$.

3.50 Estimate the value of $f(1.5, 1.5)$ in Problem 3.49 and compare the value with its actual value.

Answers to Exercises

3.1 239, 371

3.2 (a) $6h^2(h + x)$, (b) $\cos(x + 2h) - 2\cos(x + h) + \cos x$

(c) $2x + 4$ (d) $\tan^{-1}\dfrac{h}{x(x+h)}$

(e) $\dfrac{f(x+h)g(x) - g(x+h)f(x)}{g(x)\,g(x+h)}$

3.3 5.74

3.4 3.1582 **3.6** 54

3.7 $y(1) = 3, \quad y(10) = 111$ **3.8** 1.6751, 1.7082

3.10 257.259, 288.805 **3.13** 17,551

3.14 0.00000001 **3.15** 0.783172

3.17 0.3165 **3.18** 111.9152

3.19 6.7531 **3.20** 0.9856

3.21 0.9856 **3.22** 1.6858

3.23 16.25 **3.24** 0.1987

3.25 0.9763 **3.26** 3251

3.28 $\frac{1}{3}(11x^2 - 3x + 7)$ **3.29** 16

3.30 0.0239 **3.31** $-\frac{1}{15}x^3 - \frac{3}{20}x^2 + \frac{241}{60}x - \frac{39}{10}$

3.34 $-\dfrac{1}{2(x+1)} + \dfrac{1}{2(x-1)} + \dfrac{1}{x-2}$ **3.36** $\dfrac{-(a+b)}{a^2b^2}, \dfrac{ab+bc+ca}{a^2b^2c^2}$

3.37 1454 **3.40** $|y(x) - L_2(x)| = 6.415 \times 10^{-8}$

3.42 1.435081 **3.43** $1 + x^2 + x^3$

3.44 54 **3.45** 1.55

3.46 (a) **3.47** (d)

3.48 (a), (d), (f), (h) **3.49** 1.5

3.50 Proceed as in Problem 3.49.